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Term 7- Sept 2025

Nanoelectronics and Technology (01.119/99.503)-Week 9 Class 2

Shubhakar K
13-Nov- 2025

Email: shubhakar@sutd.edu.sg
Office location: 1.602-26

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Outline

- Device level characterization
- Nanoscale characterization
 - Scanning tunneling microscopy (STM)
 - Atomic force microscopy (AFM)

Probe station Electrical Characterization

- MOSFET electrical characterization at the device level involves measuring both **static** (DC current-voltage) and **dynamic** (capacitance, switching time) characteristics to understand its performance in different operating regions.
- These measurements confirm performance specifications and assess device quality

Characterization typically involves:

1. **I-V Curve Measurement:** Plotting the drain current (I_D) versus drain-source voltage (V_{DS}) (output characteristics) and I_D versus gate-source voltage (V_{GS}) (transfer characteristics) using a curve tracer or SMU instruments.

Probe station Electrical Characterization

Characterization typically involves:

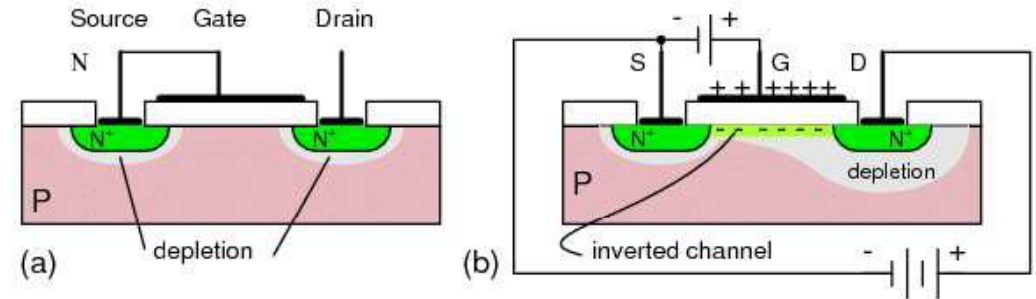
1. **I-V Curve Measurement:** Plotting the drain current (I_D) versus drain-source voltage (V_{DS}) (output characteristics) and I_D versus gate-source voltage (V_{GS}) (transfer characteristics) using a curve tracer or SMU instruments.
2. **C-V Measurement:** Measuring capacitance as a function of voltage, often using a capacitance meter or LCR meter, to extract capacitance values and interface trap densities.
3. **Specialized Tests:** Measuring specific performance metrics like breakdown voltage, leakage currents.

These measurements are crucial for design validation, failure analysis (FA), process monitoring, and ensuring a device is suitable for its intended application.

Device Level Characterization

Conventional Device level Electrical Analysis

- I-V Characterization, I-t Characterization
- C-V characterization
- Breakdown analysis at the device level



- Drawback: It is difficult to characterize the materials and device locally at the nanometer scale
- Overall average effect

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Probe station Electrical Characterization



www.keithley.com

Introduction to device characterization

Electrical Characterization:

Precision DC Current-Voltage (I-V)

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Probe station Electrical Characterization



Keithley Model 4200-SCS

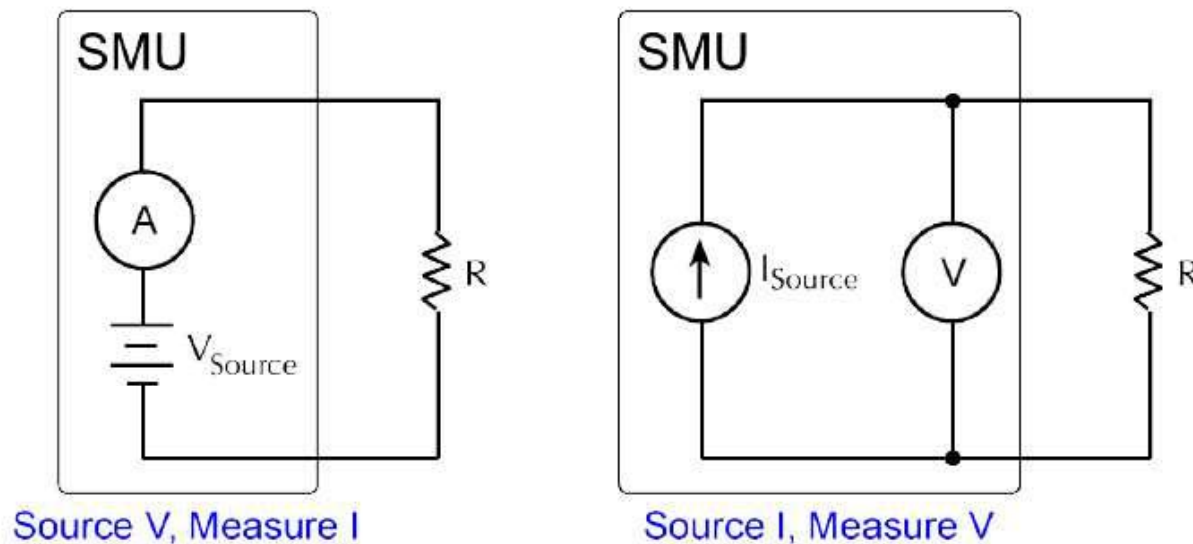
The 4200 is a complete system that integrates everything together in an easy to use system



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Probe station Electrical Characterization

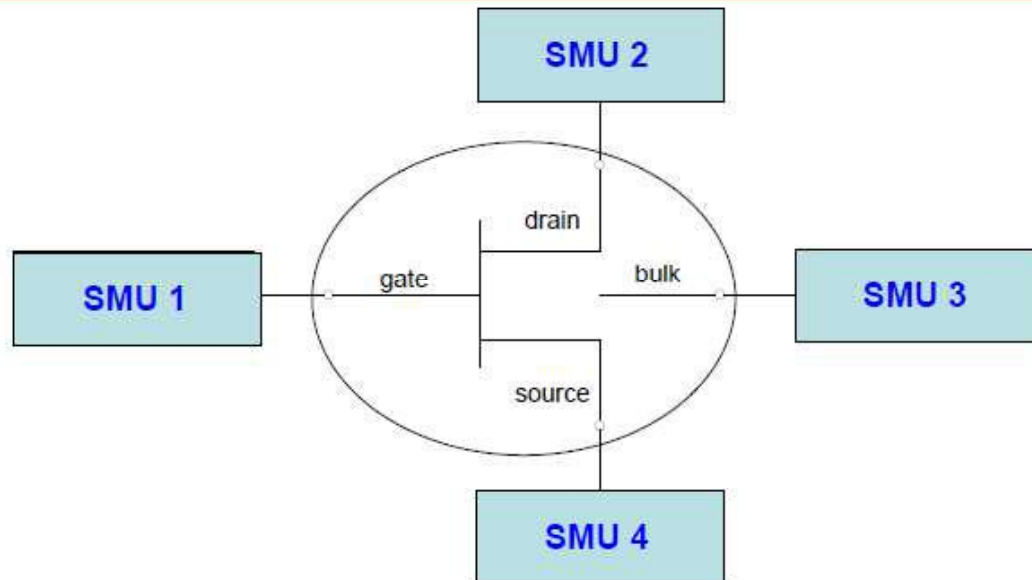
SMU stands for Source Measure Unit. An SMU is a single unit that can source and measure both current and voltage.

**KEITHLEY**

A GREATER MEASURE OF CONFIDENCE

Probe station Electrical Characterization

Multiple SMUs



Devices with multiple terminals usually required several SMUs for IV characterization of the device. In this example, the 4-terminal MOSFET requires 3 or 4 SMUs for complete testing.

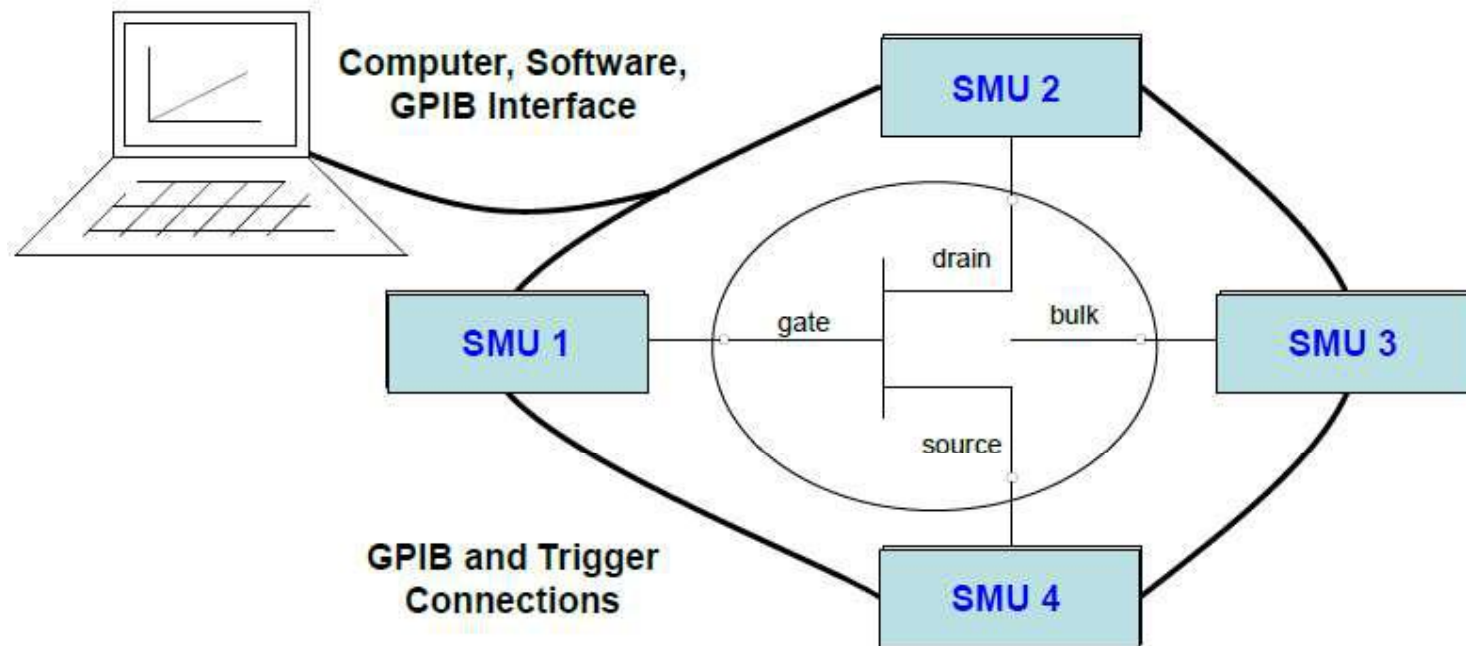
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Probe station Electrical Characterization

Automating Multiple SMUs

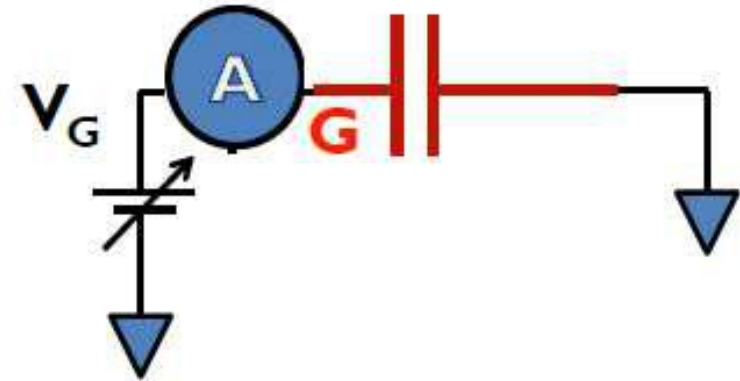
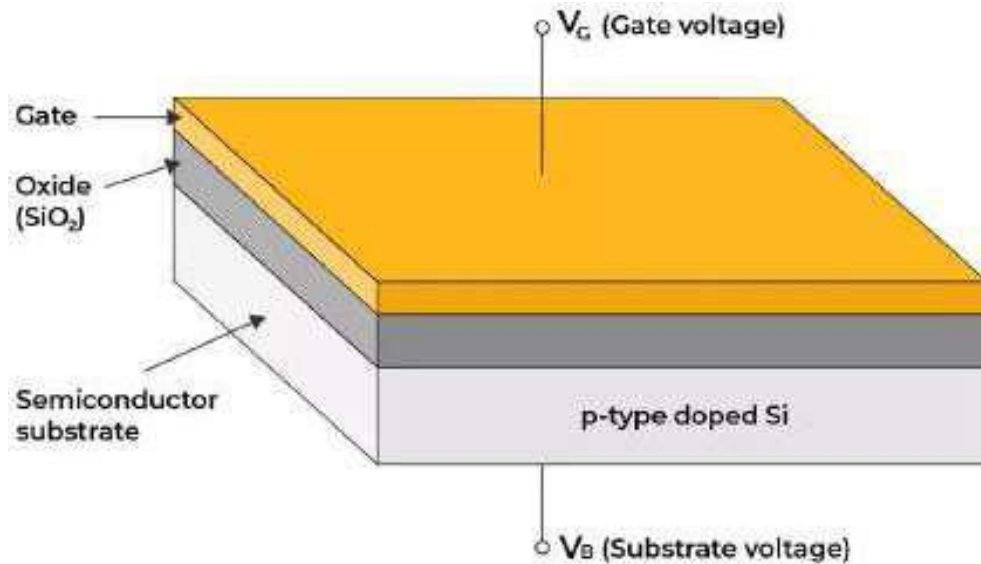
Automating and testing of multi-terminal devices requires the user to create a highly complex system involving programming of multiple instruments, precise triggering, graphics, analysis, etc.



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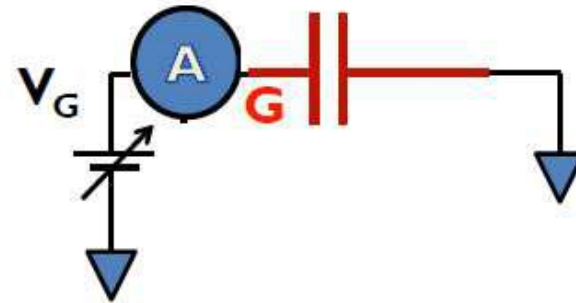
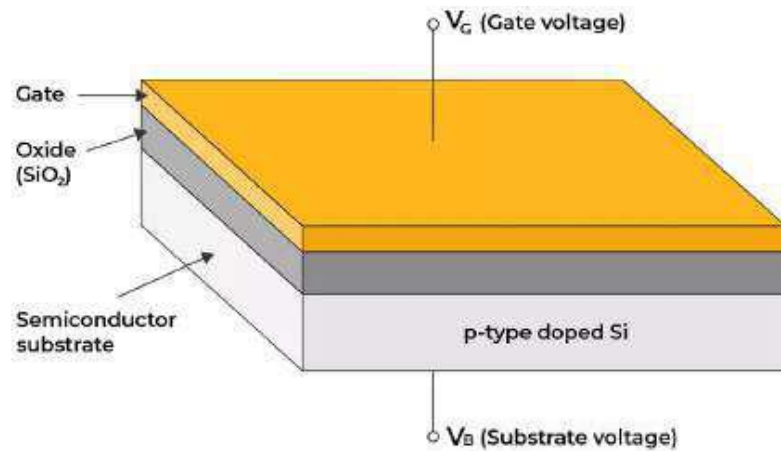
Gate dielectric characteristics



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Failure characteristics- how do we study



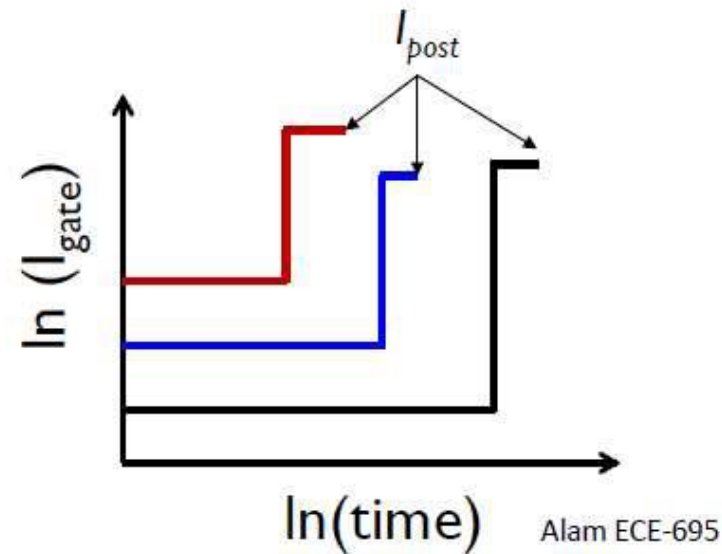
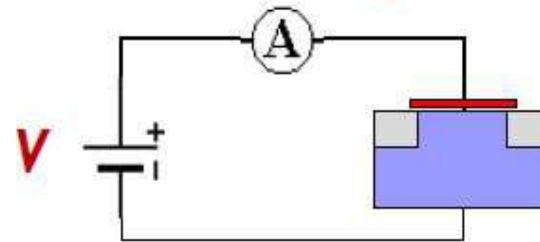
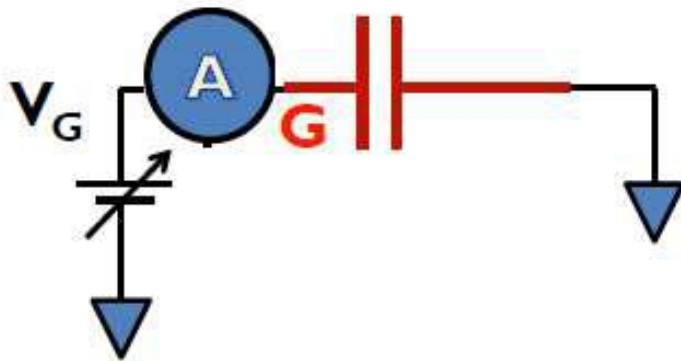
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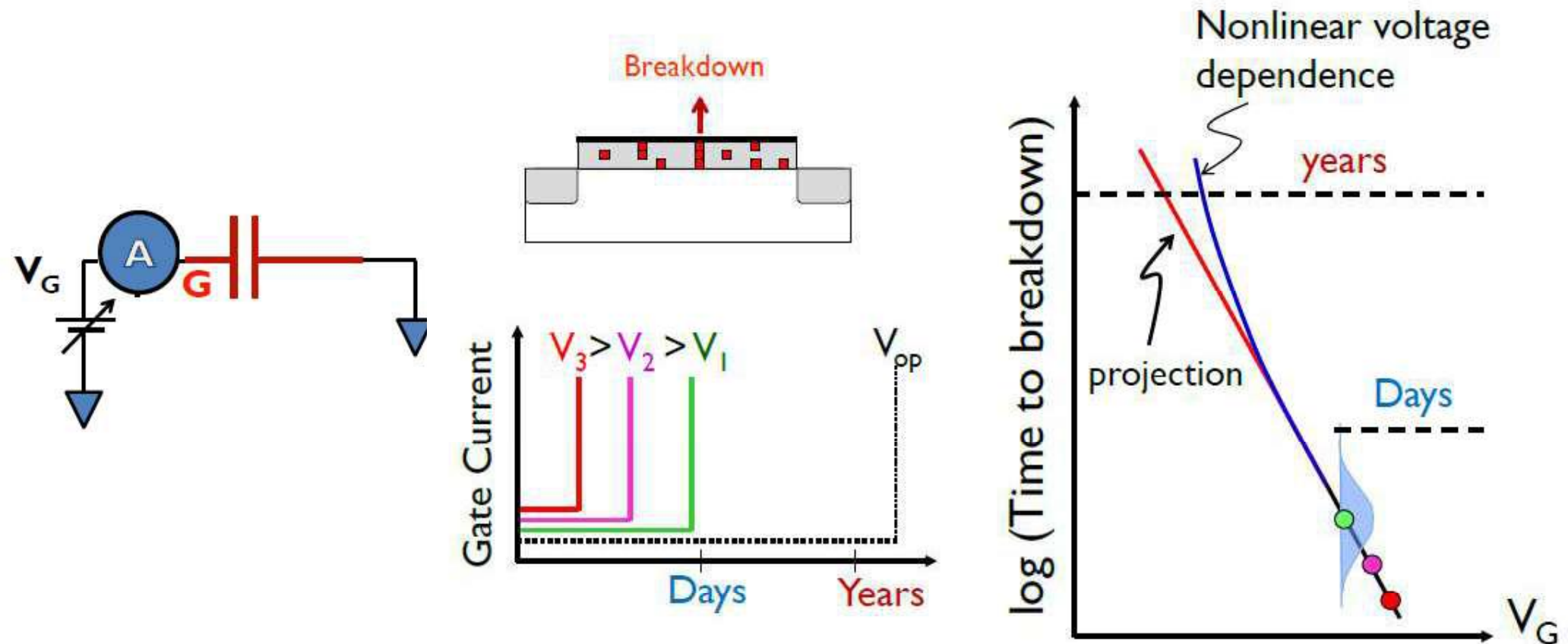
Gate dielectric characteristics

Const. Voltage Stress

Constant Voltage Stress



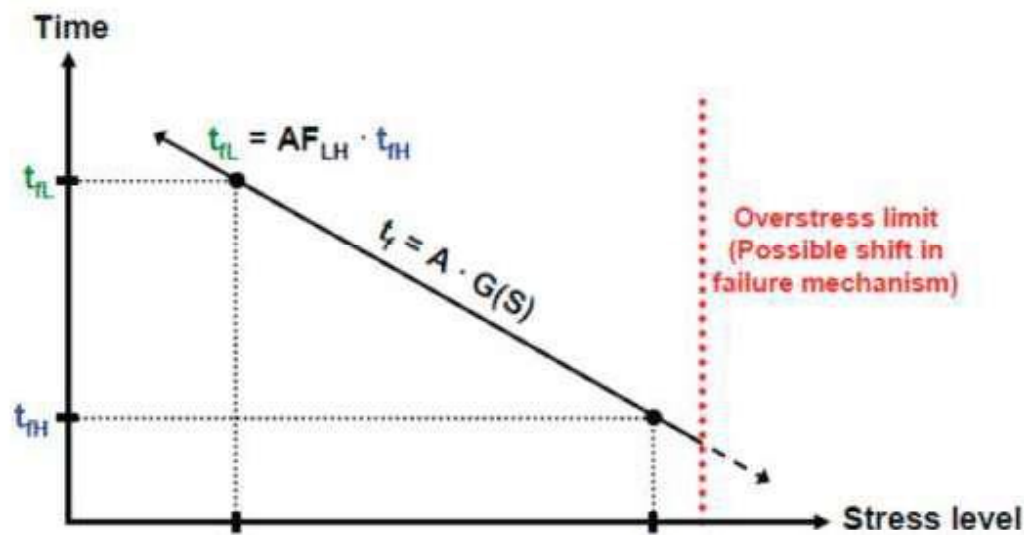
Gate dielectric characteristics -Reliability



Highly accelerated voltage dependence

Gate dielectric characteristics -Reliability

How to study reliability of the device (Dielectric)- Accelerated stressing

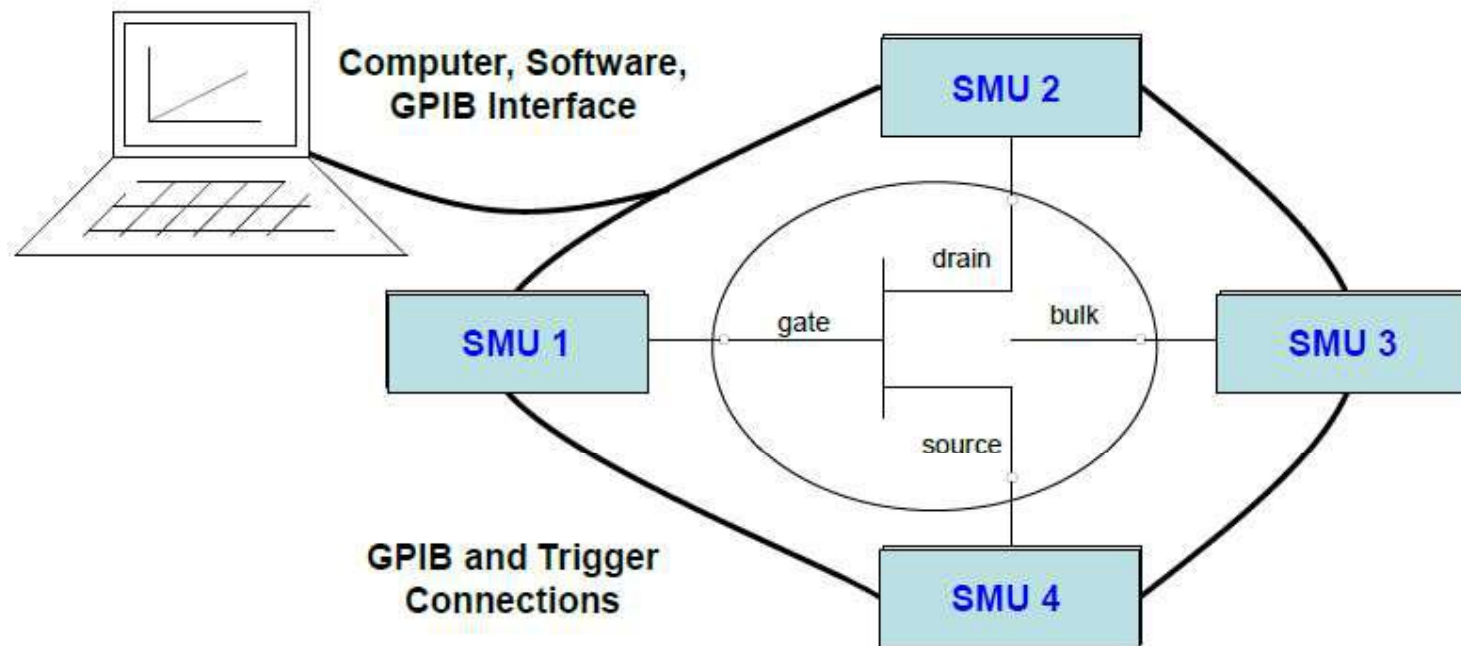


<https://www.electronic.se/2015/03/09/accelerated-life-testing-classic-or-calt/>

Probe station Electrical Characterization

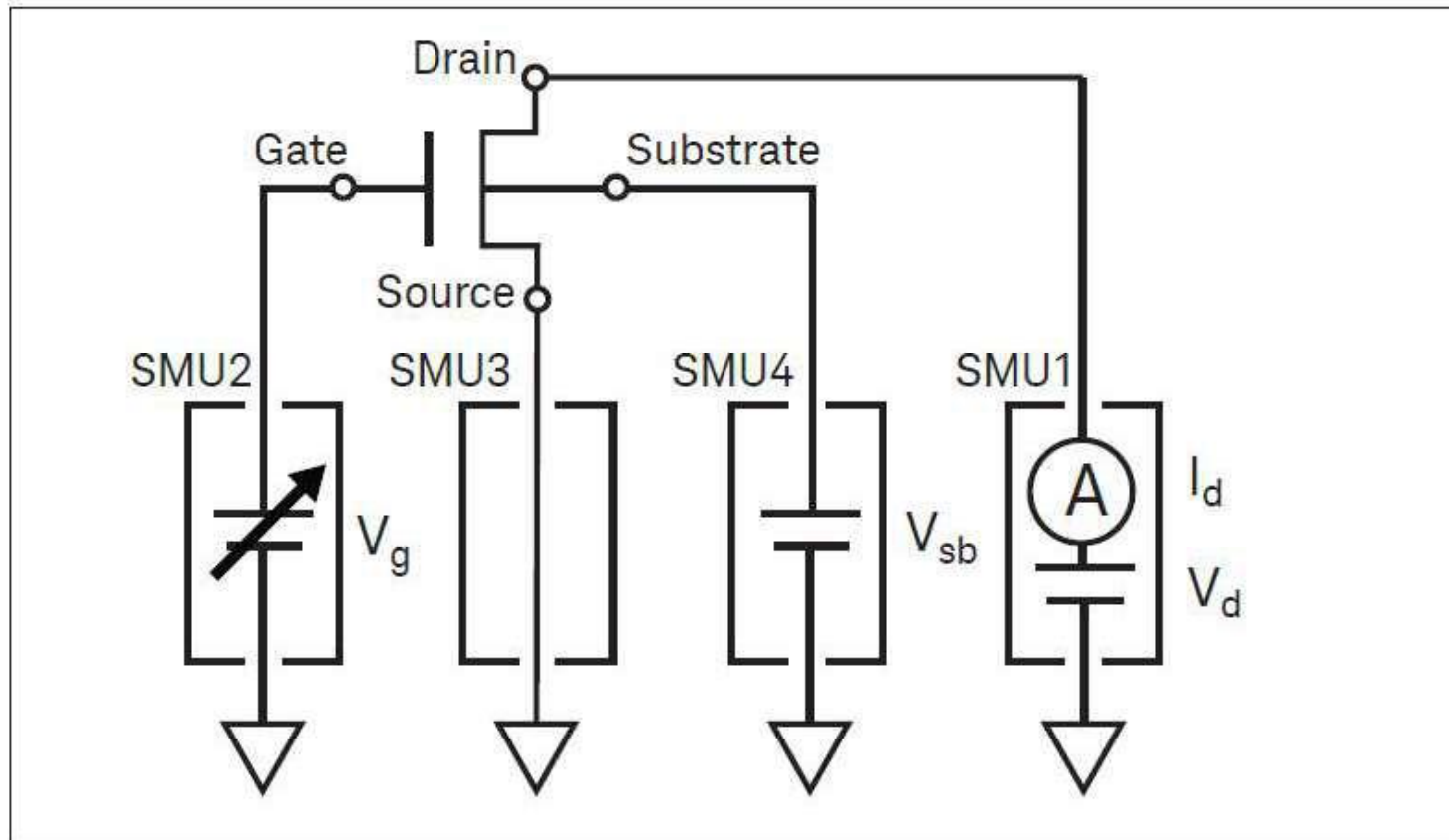
Automating Multiple SMUs

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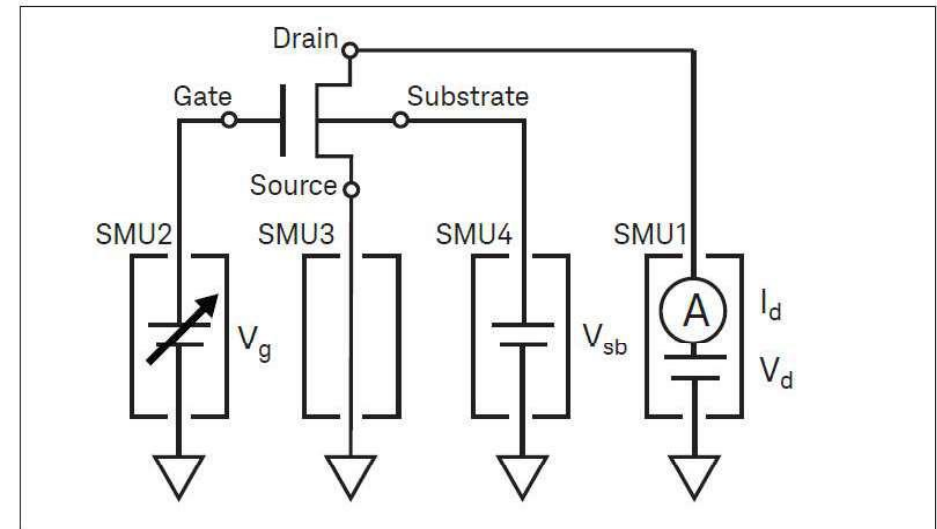
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Probe station Electrical Characterization-NMOS



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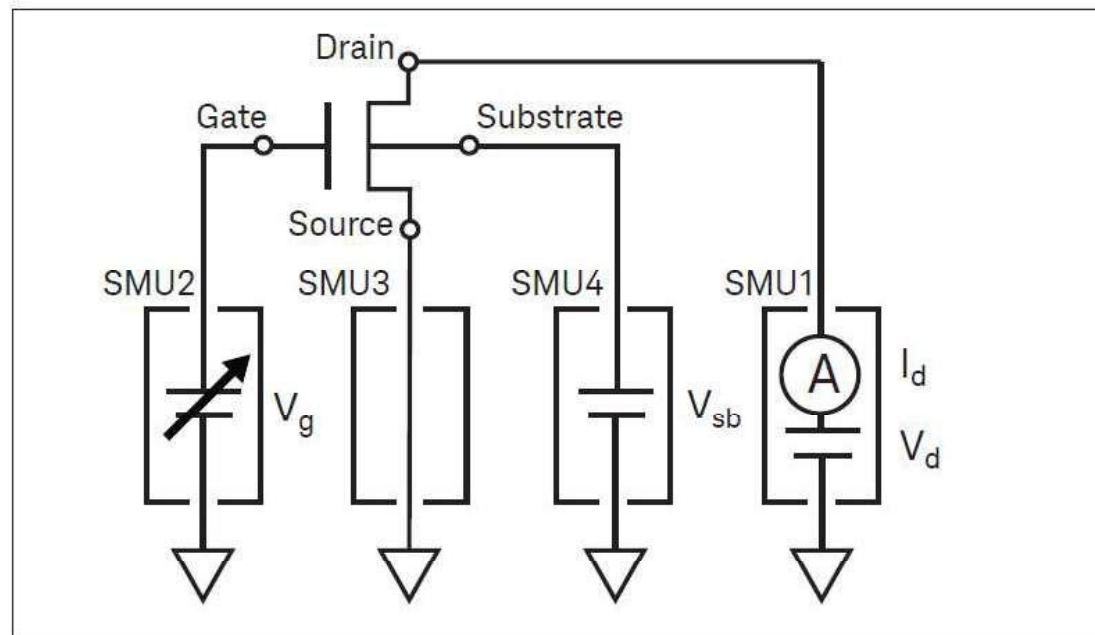
Probe station Electrical Characterization-NMOS



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Probe station Electrical Characterization-NMOS

MOSFET electrical characterization involves two primary methods: **Current-Voltage (I-V) characterization** for static behavior and **Current-Time (I-Time) or transient analysis** for dynamic/switching behavior.

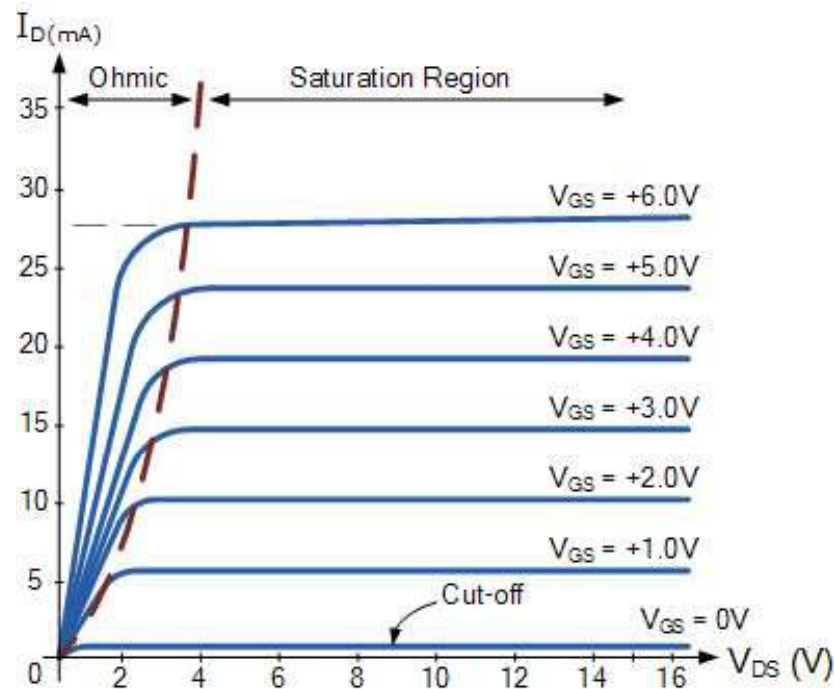


Probe station Electrical Characterization-MOSFET

- **Output Characteristics (I_D vs. V_{DS}):** Measures the drain current (I_D) as a function of the drain-source voltage (V_{DS}) for different constant gate-source voltages (V_{GS}). These curves show the device's behavior in the linear and saturation regions.
- **Transfer Characteristics (I_D vs. V_{GS}):** Measures the drain current (I_D) as a function of the gate-source voltage (V_{GS}) at a constant V_{DS} . This is used to determine the **threshold voltage (V_{th})**, the minimum gate voltage required to create a conductive channel.
- **Leakage Currents:** Characterization includes measuring parasitic currents like gate leakage current (I_{GSS}) and drain cut-off current (I_{DSS}) to assess device quality and off-state performance.
- **Breakdown Voltage (V_{BR}):** Determines the maximum voltage the device can block between the drain and source before experiencing breakdown.

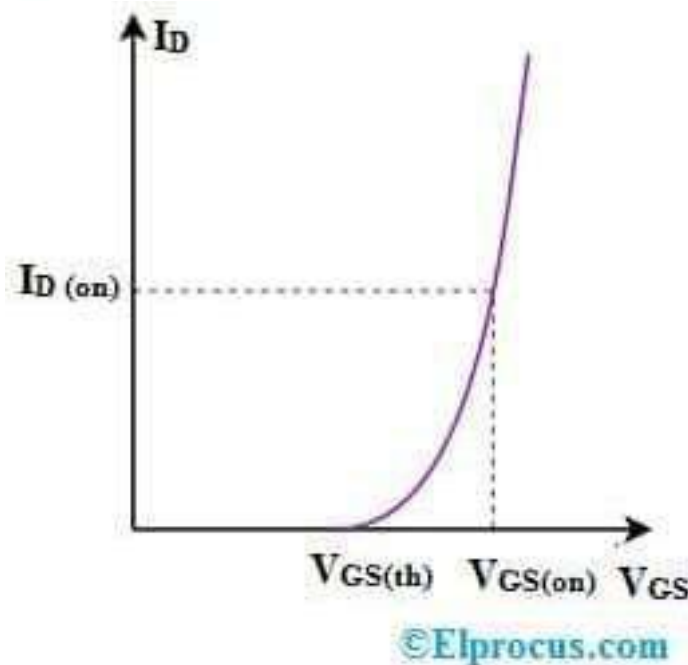
Probe station Electrical Characterization-MOSFET

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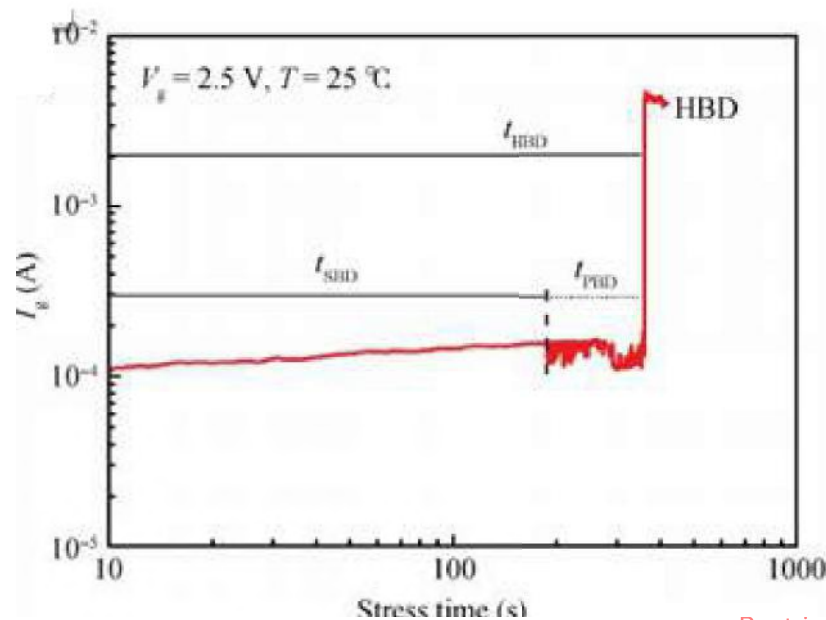
Probe station Electrical Characterization-MOSFET

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Probe station Electrical Characterization-MOSFET

- **Leakage Currents:** Characterization includes measuring parasitic currents like gate leakage current (I_{GSS}) and drain cut-off current (I_{DSS}) to assess device quality and off-state performance.
- **Breakdown Voltage ($V_{(BR)}$):** Determines the maximum voltage the device can block between the drain and source before experiencing breakdown.



<https://m.blog.naver.com/framkang/220718595715>

Nanoscale Characterization

Nanoscale level characterization presents unique physical, electrical and chemical properties compared to their respective particles at higher scales (device level).

- Scanning Tunneling Microscopy (STM)
 - Scanning Tunneling Microscopy (STM)
 - Atomic Force Microscopy (AFM)
- Electron Microscopy (TEM)
 - Transmission Electron Microscopy (TEM)
 - Scanning Electron Microscopy (SEM)
- Spectroscopic Techniques
 - X-ray Photoelectron Spectroscopy (XPS)
 - Energy-dispersive X-ray spectroscopy (EDX)

Nanoscale Characterization

Nanoscale level characterization presents unique physical, electrical and chemical properties compared to their respective particles at higher scales (device level).

- **Small structural variations (at nanoscale) create large electrical variations**
 - Missing Si atoms → threshold voltage shifts
 - Small roughness → mobility degradation
- We need **high-resolution techniques** to measure:
 - interface roughness
 - gate oxide thickness
 - line-edge roughness
- Most reliability and failures in advanced CMOS are **nanometer-scale phenomena**
- Without nanoscale characterization, nanoelectronics cannot be understood, optimized, or scaled effectively.

- Scanning Tunneling Microscopy (STM)
- Atomic Force Microscopy (AFM)
- Transmission Electron Microscopy (TEM)
- Scanning Electron Microscopy (SEM)



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SEM, AFM & STM

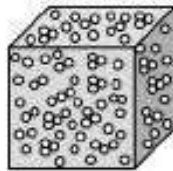


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Ultra-High Vacuum (UHV) condition

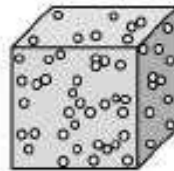
- UHV conditions are created by pumping the gas out of a UHV chamber.
- As the vacuum is the absence of material, what is in fact measured is the pressure of residual gas in the chamber. Vacuum represents particle density lower than the atmospheric pressure.
- UHV conditions are integral to scientific research and Surface science experiments often require a chemically clean sample surface with the absence of any unwanted adsorbates.

Rough Vacuum
 $1 \text{ atm} - 10^{-3} \text{ Torr}$



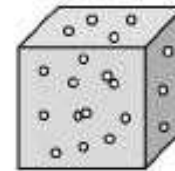
$1 \cdot 10^{-3} \text{ Torr}$
 $4 \cdot 10^{13} \text{ atom/cm}^3$

High Vacuum
 $10^{-3} \text{ Torr} - 10^{-8} \text{ Torr}$



$1 \cdot 10^{-8} \text{ Torr}$
 $4 \cdot 10^{10} \text{ atom/cm}^3$

Ultra High Vacuum
 $10^{-8} \text{ Torr} - 10^{-12} \text{ Torr}$



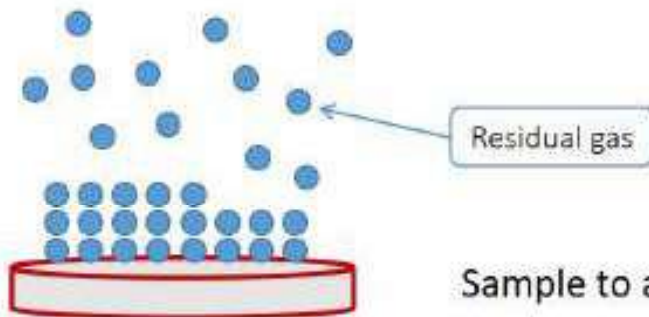
$1 \cdot 10^{-11} \text{ Torr}$
 $4 \cdot 10^5 \text{ atom/cm}^3$

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Ultra-High Vacuum (UHV) condition

High Vacuum (HV)
from 4×10^{-7} to 4×10^{-6} Torr

1 residual gas monolayer
every **4 SECONDS**



Ultra High Vacuum (UHV)
from 7.5×10^{-12} to 7.5×10^{-11} Torr

1 residual gas monolayer
every **4 DAYS**



Hectopascal [hPa]

Torr [Torr]

1 hPa

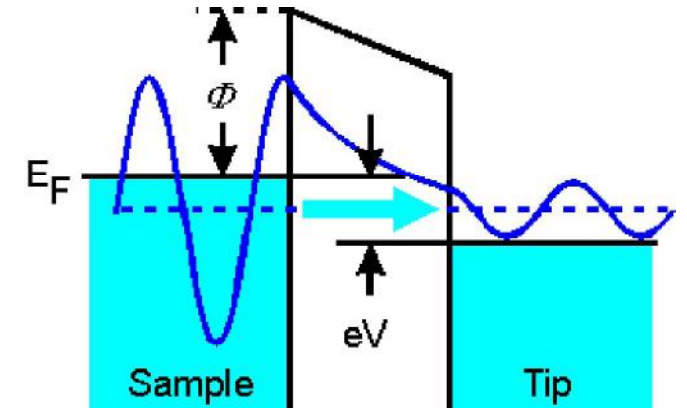
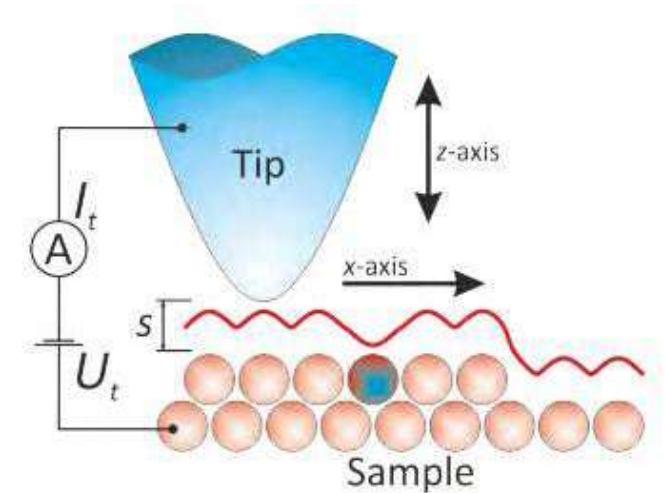
0.7500616827 Torr

- Keeps surface clean for study
- Avoids anodic oxidation

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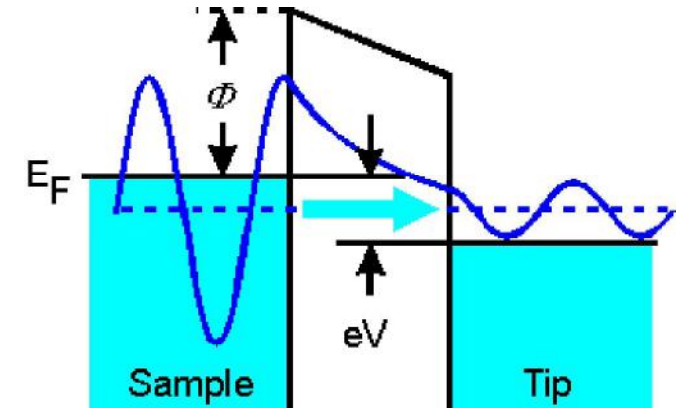
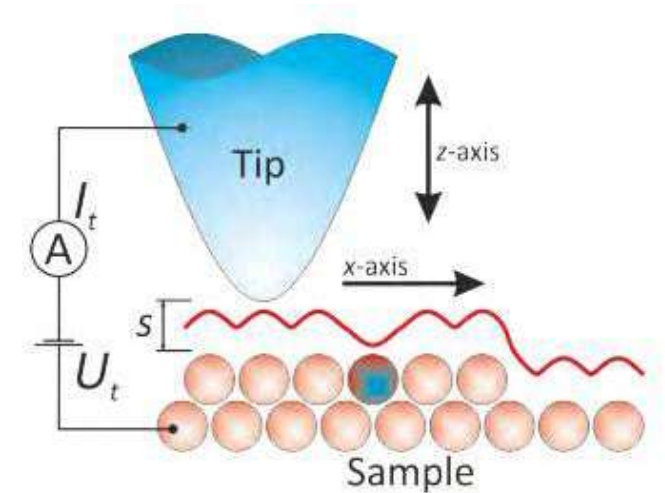
Scanning Tunneling Microscopy

- **Scanning tunneling microscope (STM)** is a type of microscope used for imaging surfaces at the atomic level.
- Its development in 1981 earned its inventors, Gerd Binnig and Heinrich Rohrer, then at IBM Zürich, the Nobel Prize in Physics in 1986.



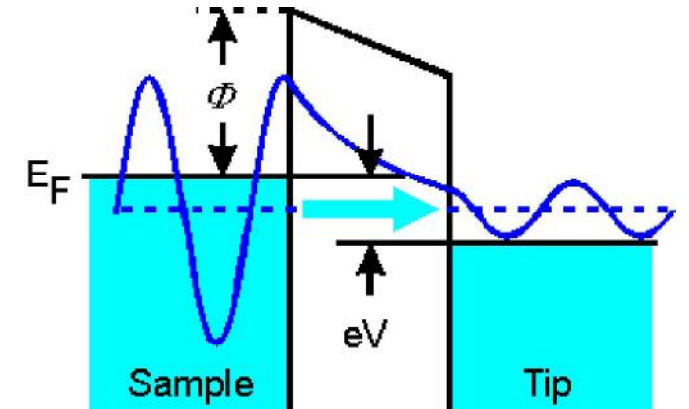
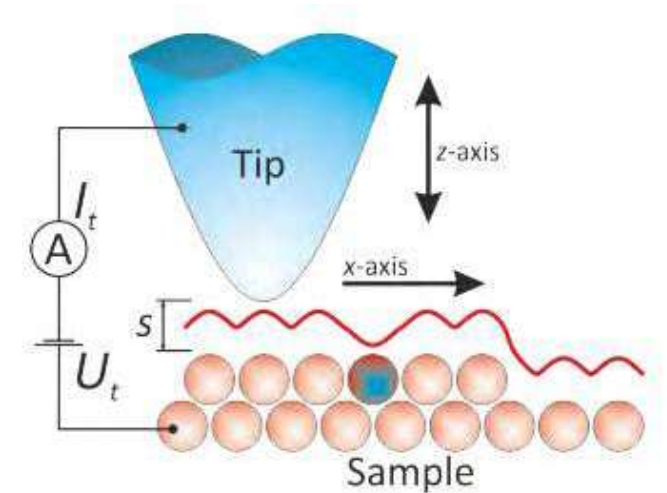
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- STM senses the surface by using an extremely sharp conducting tip that can distinguish features at sub-nanometer resolution.
- This means that individual atoms can routinely be imaged and manipulated.
- STM operated at ultra-high vacuum.
- STM is based on the concept of quantum tunneling.



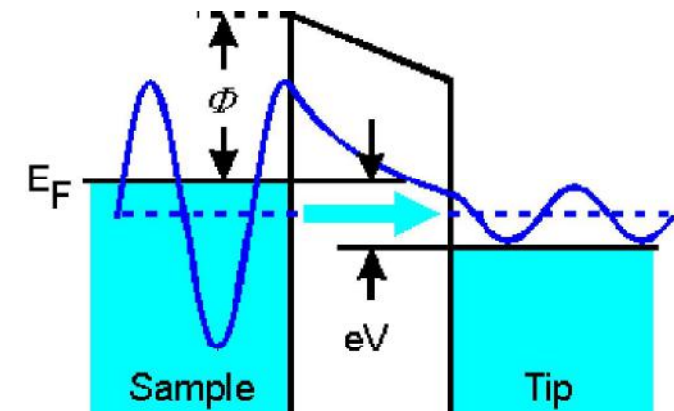
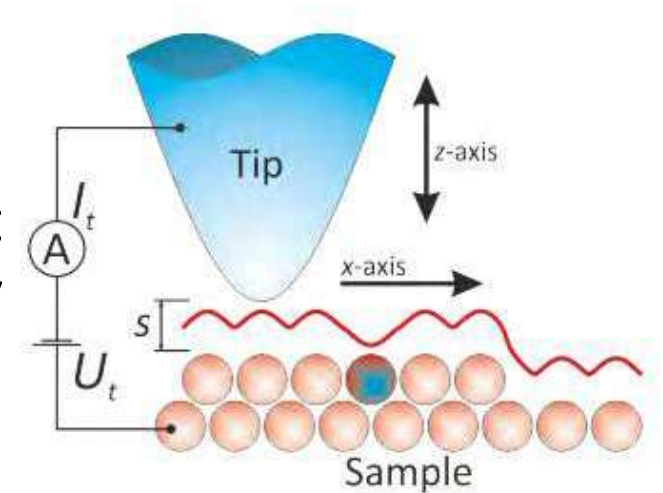
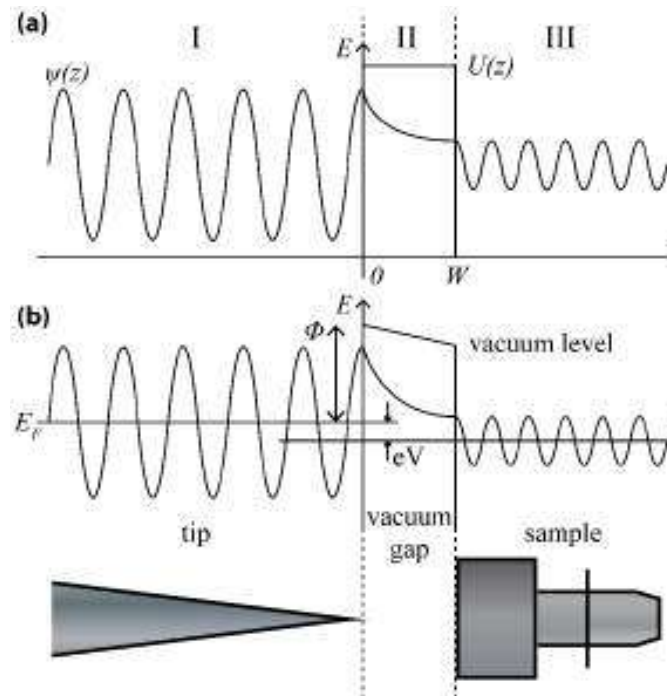
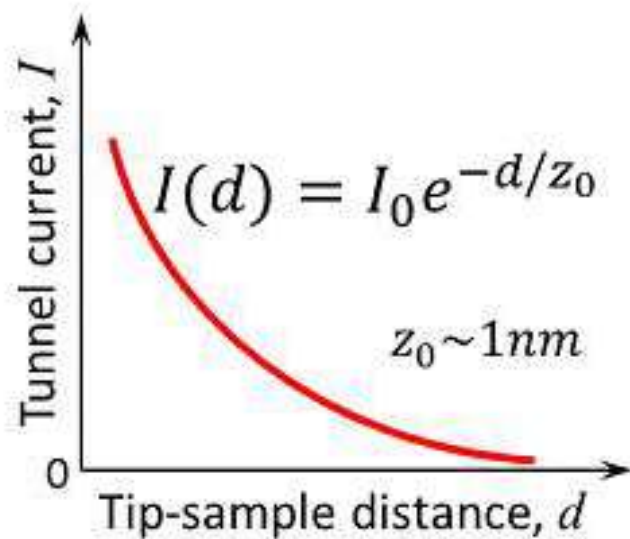
Scanning Tunneling Microscopy

- STM operated at ultra-high vacuum.
- STM is based on the concept of quantum tunneling.
- When the tip is brought very near to the surface to be examined, a bias voltage applied between the two allows electrons to tunnel through the vacuum separating them.
- STM can be a challenging technique, as it requires extremely clean and stable surfaces, sharp tips, excellent vibration isolation, and sophisticated electronics.



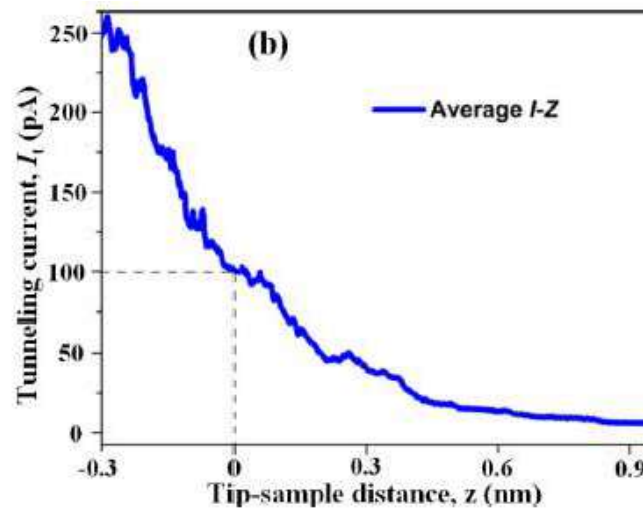
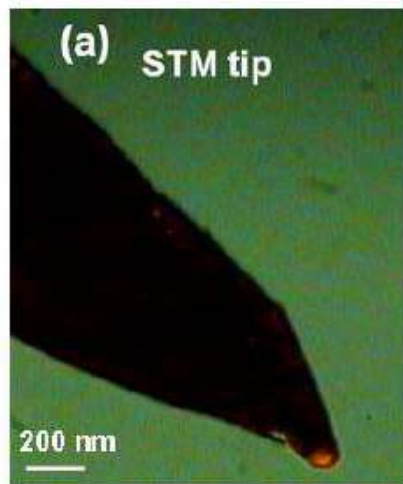
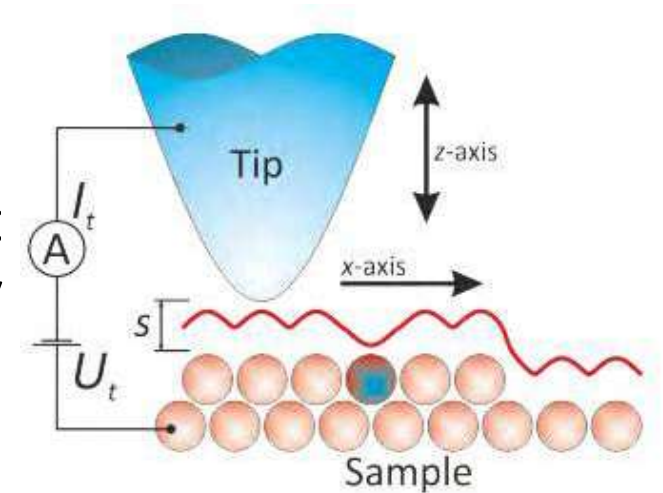
Scanning Tunneling Microscopy

- STM is based on the fact that the tunneling current between a conductive tip and sample is exponentially dependent on their separation.

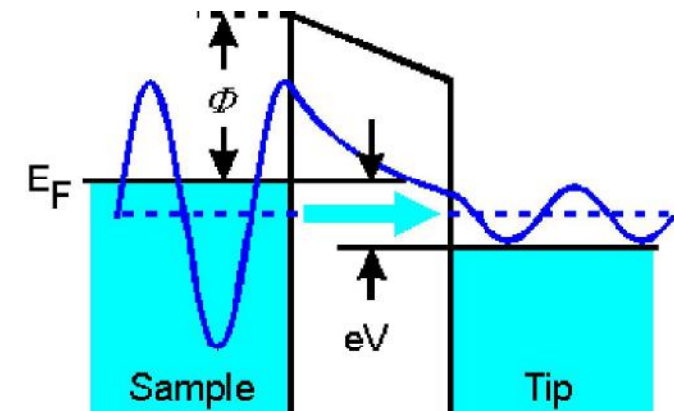


Scanning Tunneling Microscopy

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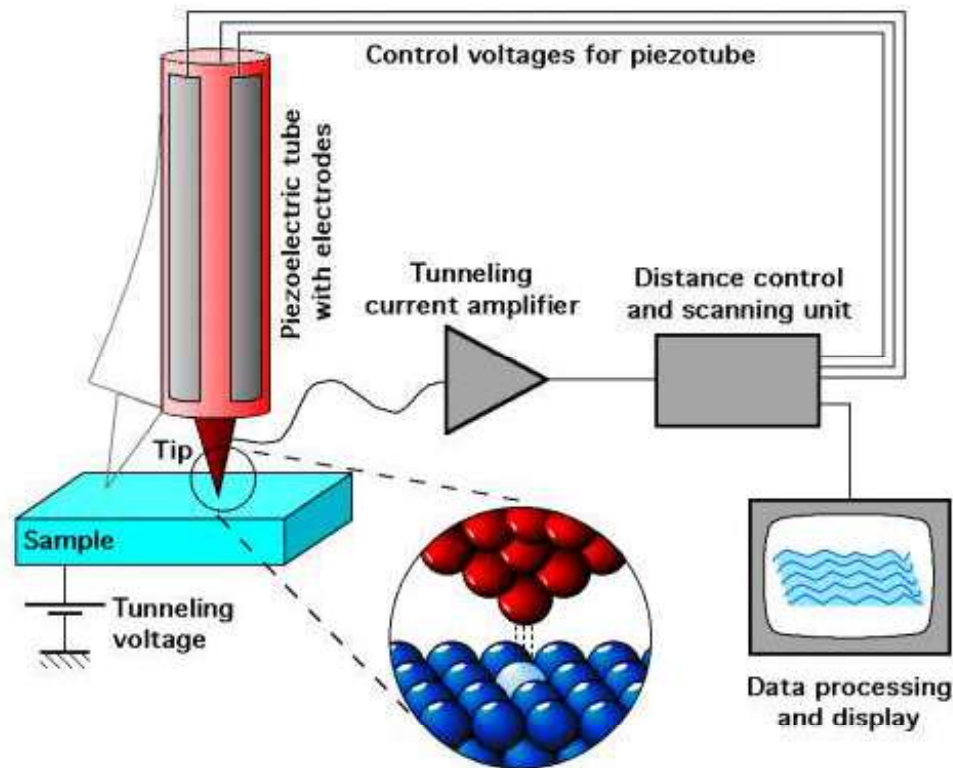


(a) Optical microscope image of the Pt-Ir STM tip, (b) I - z (average of 4-5 readings) characteristics showing tunneling probability w.r.t tip-sample distance.



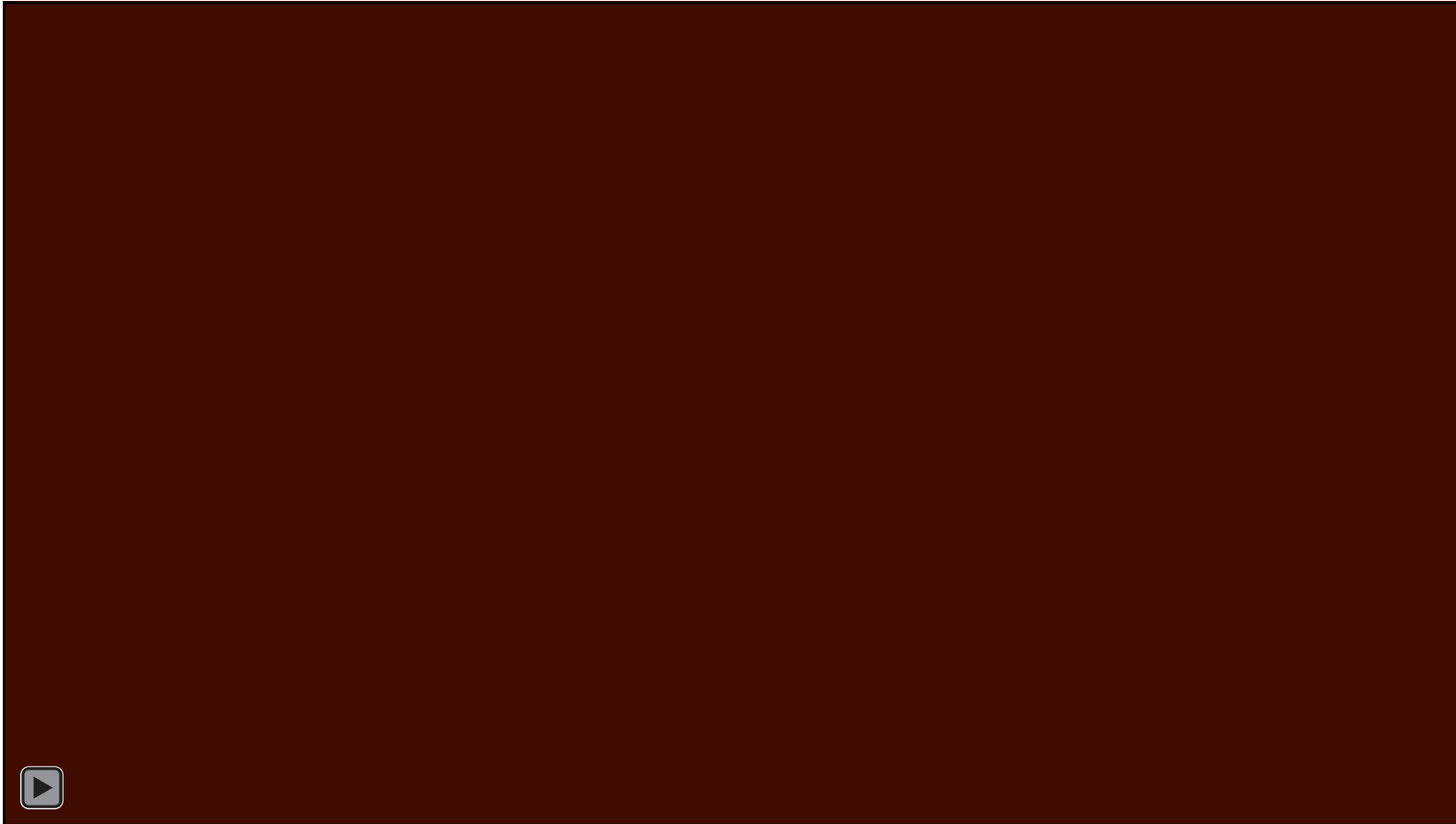
Scanning Tunneling Microscopy

- STM-tunneling of electrons from sample to tip and vice versa through vacuum
- UHV-conditions, Tip- Pt and W



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Scanning Tunneling Microscopy



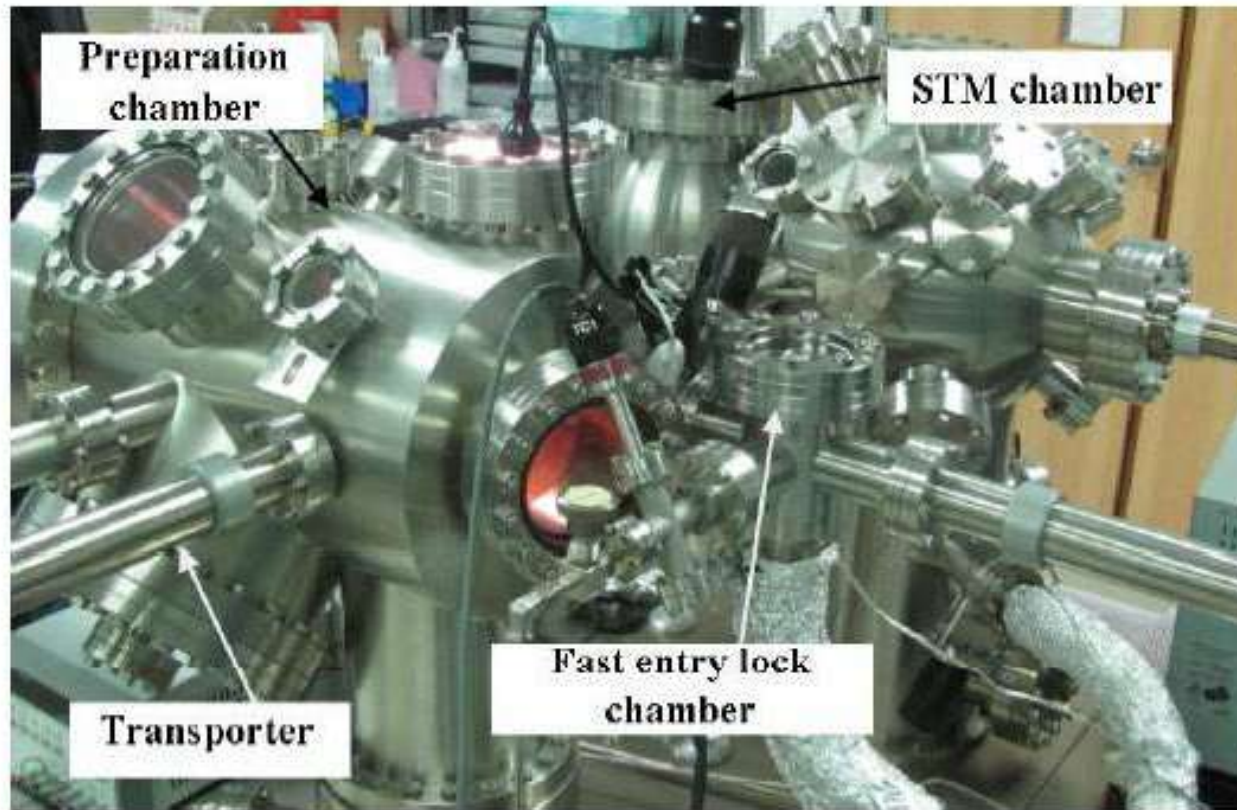
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Scanning Tunneling Microscopy

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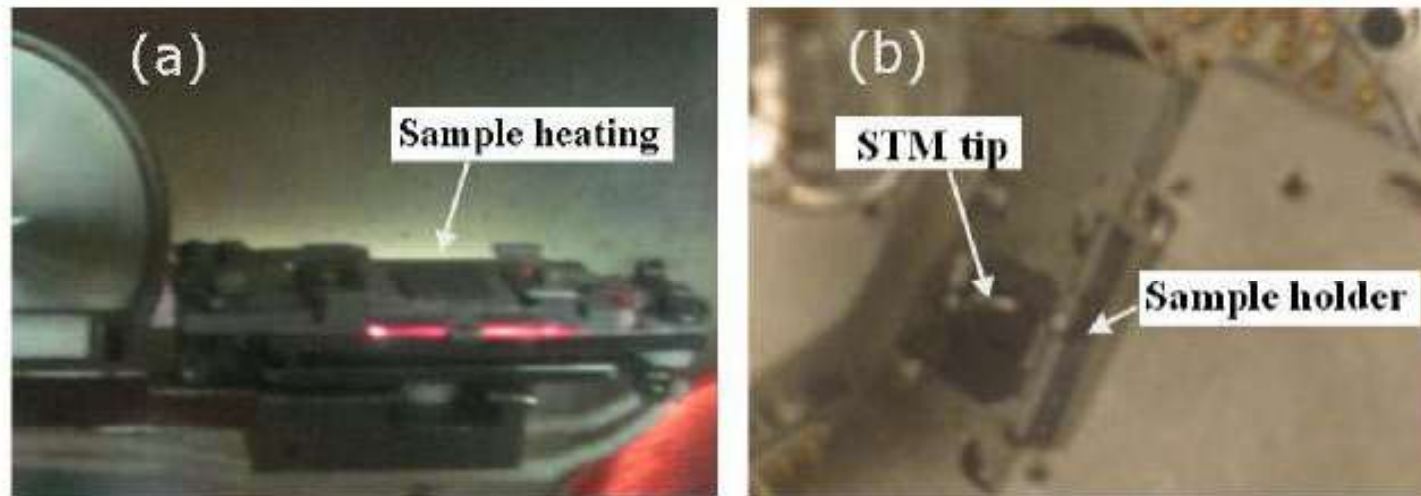


The UHV-STM system consisting of a fast entry lock chamber, a preparation chamber and a STM chamber.

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Scanning Tunneling Microscopy

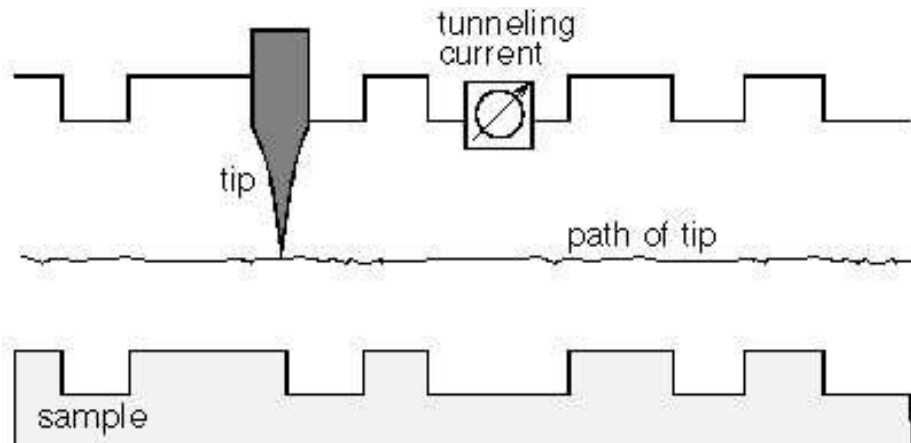


(a) Sample heating in resistive heating mode in the preparation chamber. (b) STM tip and sample holder setup inside the main STM chamber.

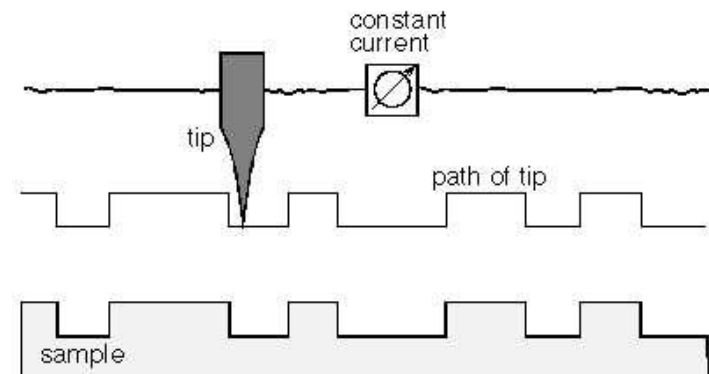
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Scanning Tunneling Microscopy- Operating modes

- Constant Height
- Constant current



Constant Height

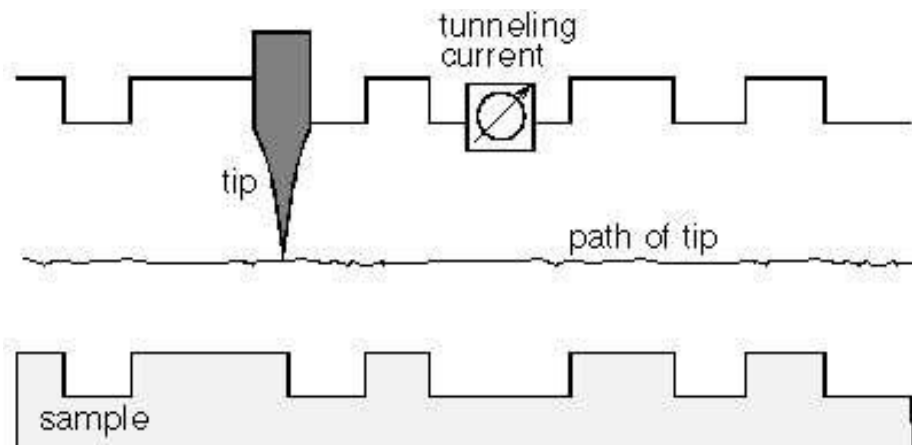


Constant current

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Scanning Tunneling Microscopy- Operating modes

- Constant Height

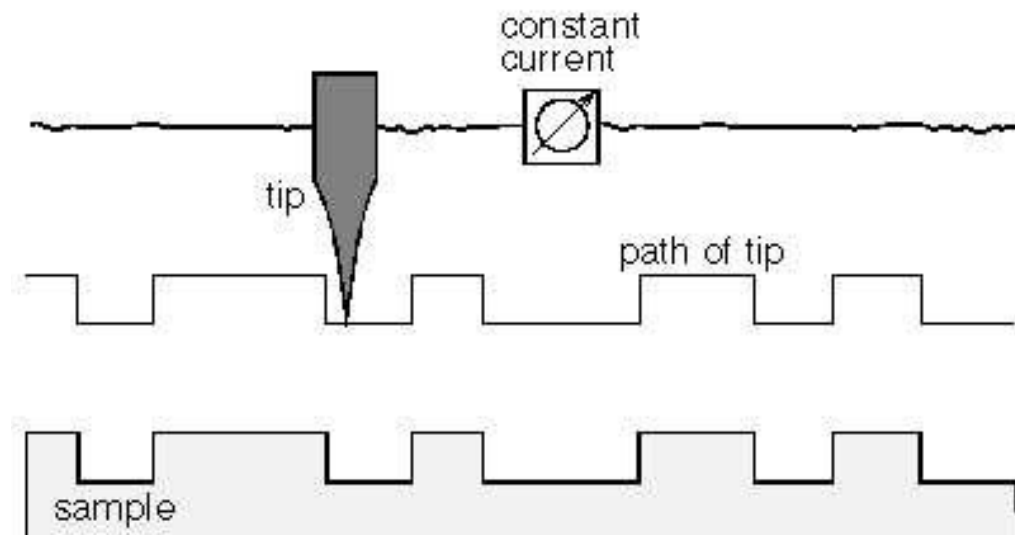


Constant Height

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Scanning Tunneling Microscopy- Operating modes

- Constant current



Constant current

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Scanning Tunneling Microscopy- Operating modes

STM Experimental procedure steps

The following experimental procedures were adopted for our STM experiments.

- (a) A Blanket HK dielectric sample with size approximately 1 cm x 1 cm was cut and subjected to surface cleaning using isopropyl alcohol and acetone.
- (b) The sample was dried with dry nitrogen and mounted on a sample holder.
- (c) The sample/tip was loaded in the FEL chamber. After the FEL chamber is pumped down to $\sim 10^{-6}$ torr by a turbo molecular pump, the valve between the FEL chamber and the preparation chamber is opened for sample/tip transfer.

Scanning Tunneling Microscopy- Operating modes

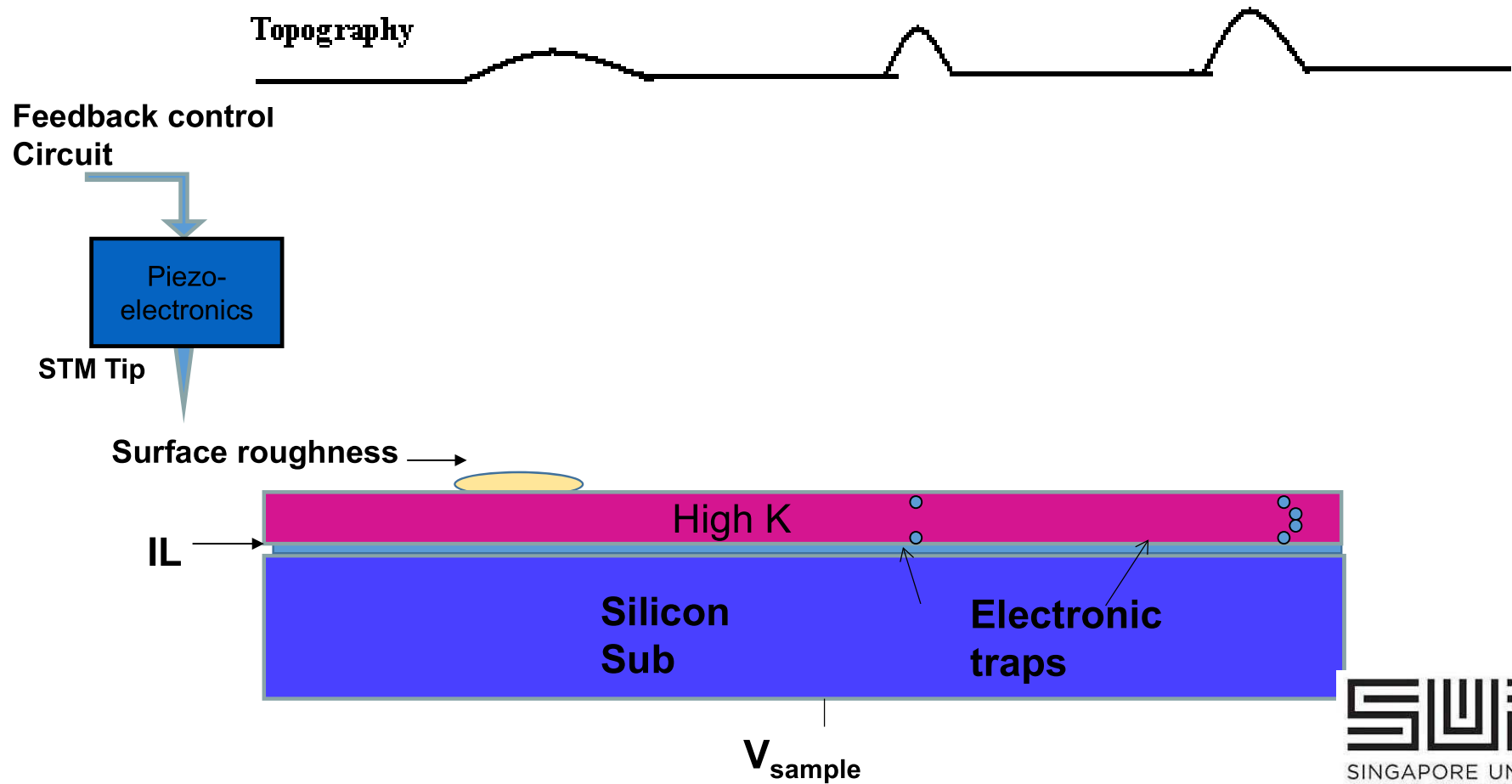
STM Experimental procedure steps

- (d) The samples were degassed at $\sim 120^\circ\text{C}$ in the preparation chamber and then transferred into the STM chamber.
- (e) Tips were transferred directly into the STM chamber.
- (f) The STM tip was brought closer to the sample surface and performed the STM operation and analysis.

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Scanning Tunneling Microscopy

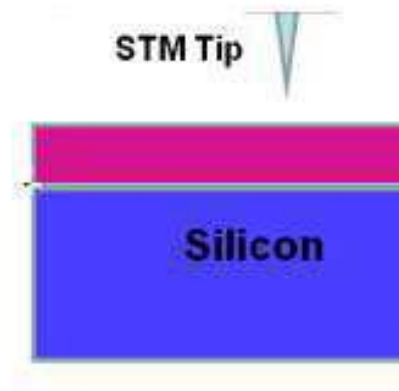
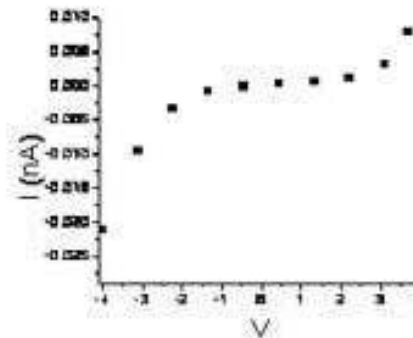
Topography → Morphology + local electronic property



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Scanning Tunneling Spectroscopy (STS)

Scanning tunneling spectroscopy (STS), an extension of scanning tunneling microscopy (STM), is used to provide information about the density of electrons in a sample as a function of their energy.

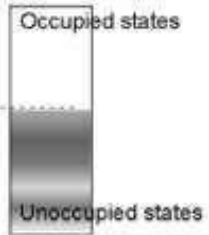
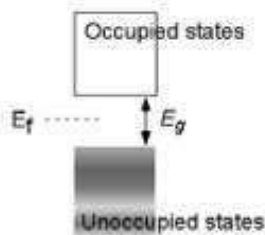
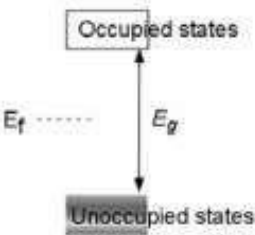
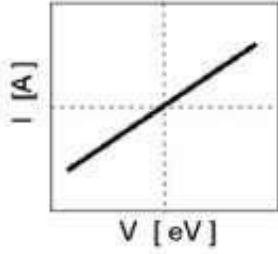
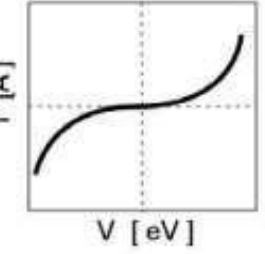
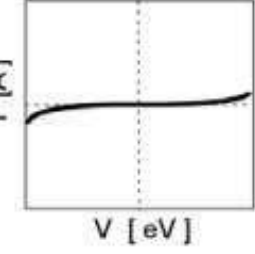
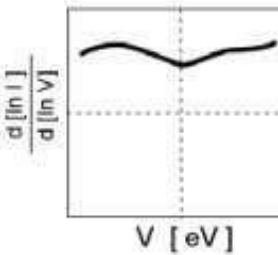
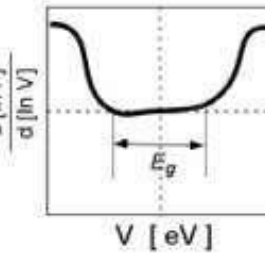
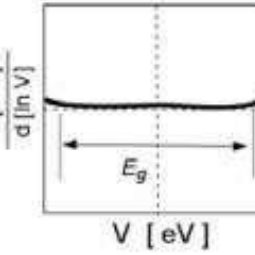


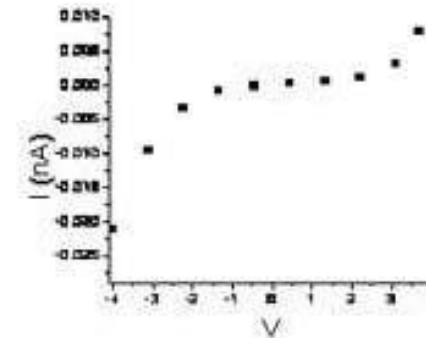
Scanning Tunneling Spectroscopy (STS)

Scanning tunneling spectroscopy (STS), an extension of scanning tunneling microscopy (STM), is used to provide information about the density of electrons in a sample as a function of their energy.

- Scanning tunneling spectrum is obtained by placing a scanning tunneling microscope tip above a particular place on the sample.
- With the height of the tip fixed, the electron tunneling current is then measured as a function of electron energy by varying the voltage between the tip and the sample.
- The change of the current with the energy of the electrons is the simplest spectrum that can be obtained, it is often referred to as an I-V curve.
- As is shown below, it is the slope of the I-V curve at each voltage (often called the dI/dV -curve) which is more fundamental because dI/dV corresponds to the electron density of states

Scanning Tunneling Spectroscopy

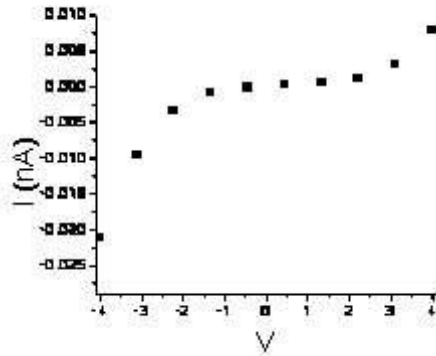
	Metal	Semiconductor	Insulator
Electronic Structure			
I-V			
dI/dV			



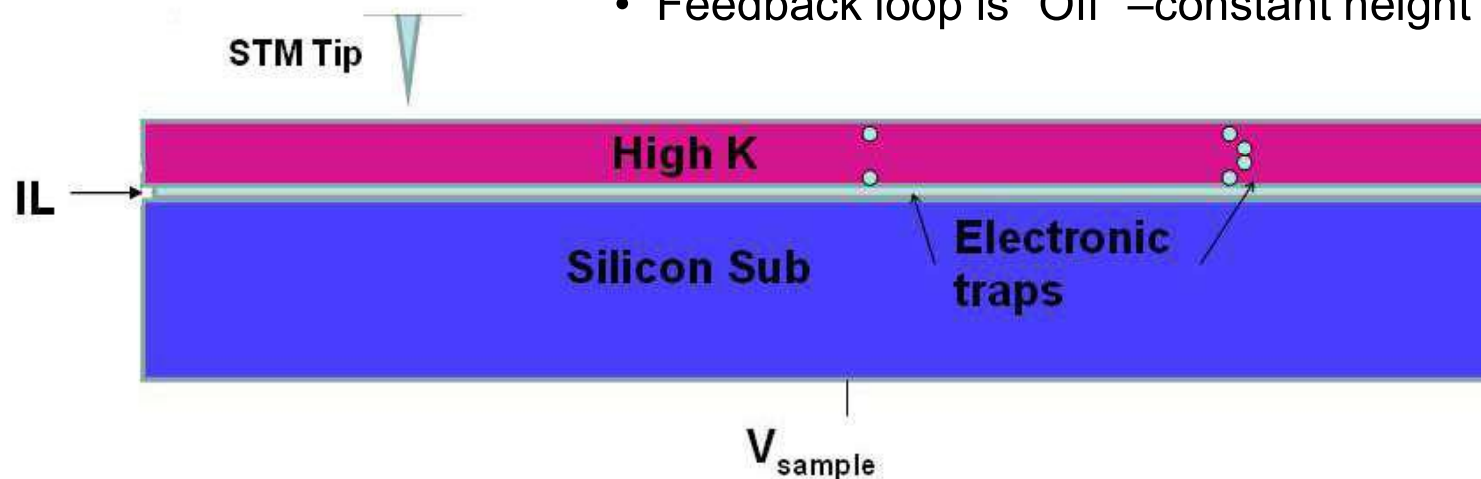
STM Tip



Current Induced Tunneling Spectroscopy

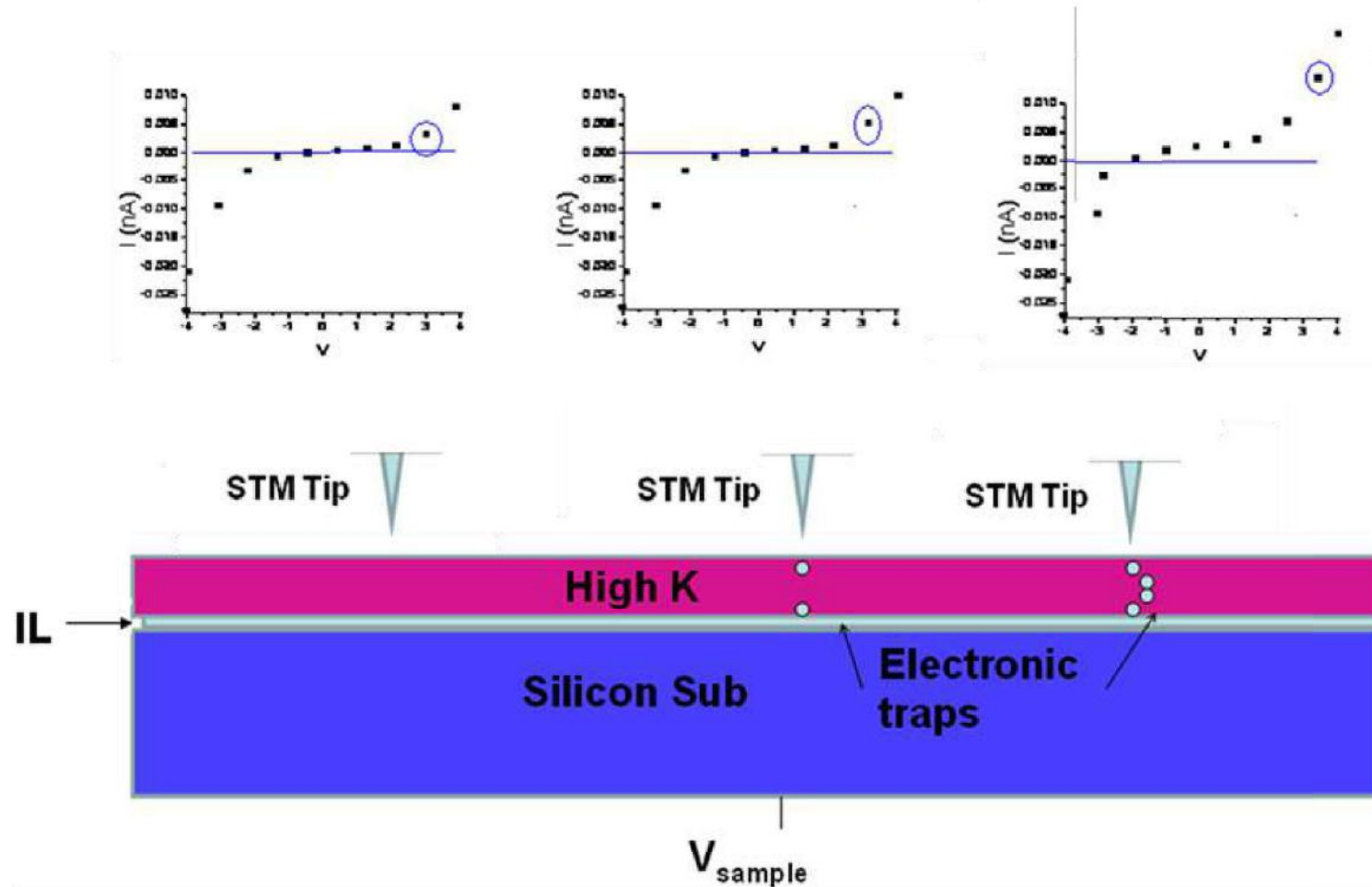


- In order to separate the electrical properties from morphology → CITS
- CITS → local I - V characteristic at every pixel in an image
- Feedback loop is “Off” –constant height mode



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Current Induced Tunneling Spectroscopy



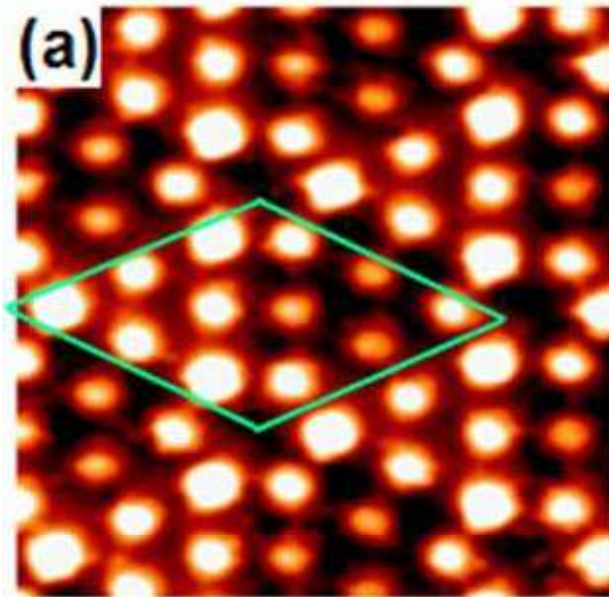
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STM Applications

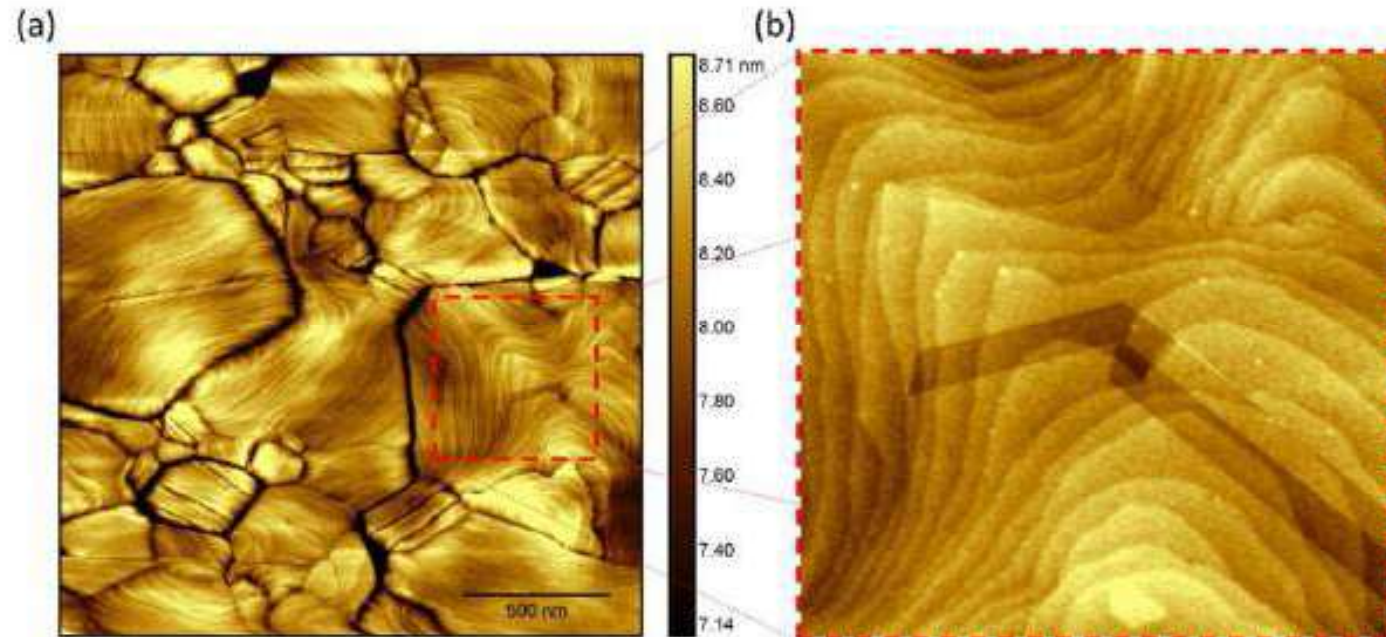
- Surface structure analysis compared to Bulk material- Topography at the atomic level/nanometer scale
- Surface structure analysis applicable to wide variety of samples
- Spectroscopy of local area of the samples- Extract local electrical characteristics of the materials

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Scanning Tunneling Microscopy Images



Silicon surface

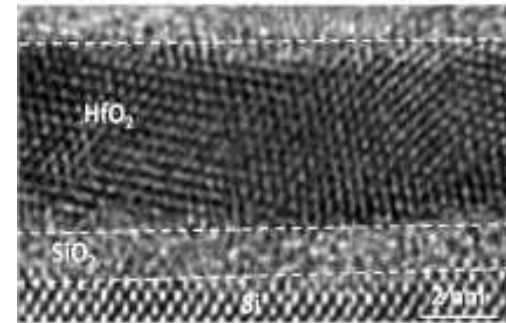
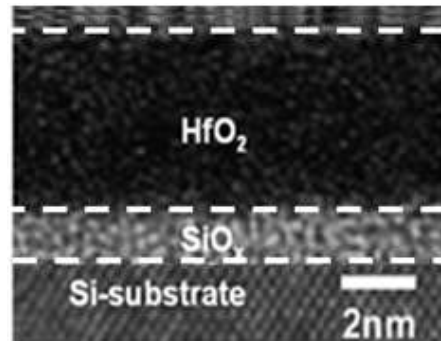


Gold surface

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STM: Effect of PDA on HfO₂ Dielectric

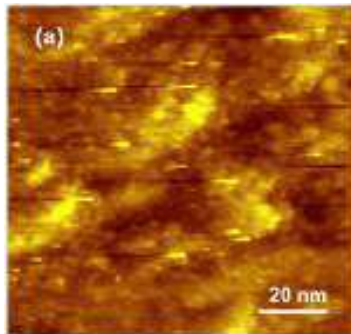
PDA-Post Deposition
Annealing



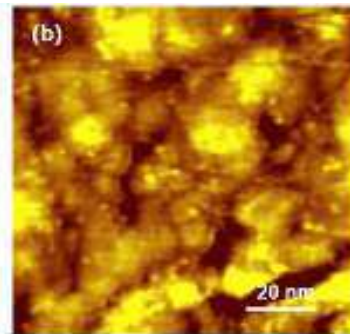
As-deposited [H. S. Jung, *Electrochem. and Sol. State Lett.*, 14, (2011)]

600°C annealed HfO₂

Topography (+3.5V, 30pA)

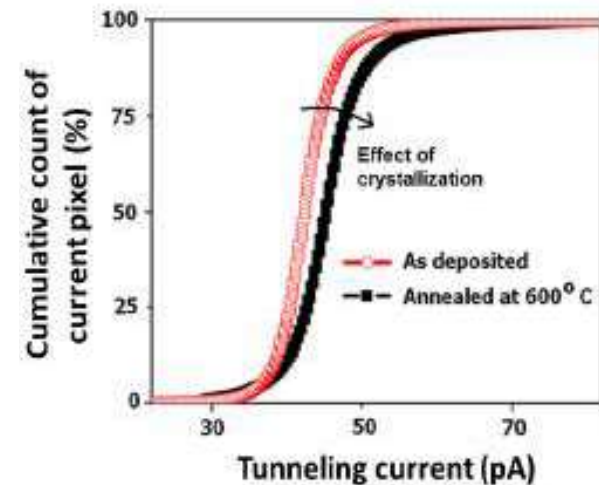


As-deposited



Annealed at 600 °C

CITS
Analysis
(+4V)



- Microstructural changes after annealing
- Increase in the surface roughness – (0.38 nm (RMS)- 0.74 nm (RMS))
- Increase in leakage current –CITS study

Key Challenges in STM

- **Sample Preparation**-High Surface Quality
- **Material Compatibility:** Some materials, such as insulators, are difficult to study with STM due to their lack of conductivity. Although variations like scanning tunneling spectroscopy (STS) and atomic force microscopy (AFM) can address this, conducting materials are preferred.

Key Challenges in STM

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- Sample Preparation-High Surface Quality
- **Material Compatibility:** Some materials, such as insulators, are difficult to study with STM due to their lack of conductivity. Although variations like scanning tunneling spectroscopy (STS) and atomic force microscopy (AFM) can address this, conducting materials are preferred.
- **Handling Sensitive Samples:** Some materials, particularly biological samples or organic molecules, can be damaged under the high-energy electron beam used in STM, requiring careful handling or cryogenic conditions.
- **Tip-Sample Interaction (Tip Stability):** The sharpness and cleanliness of the STM tip are crucial for high-resolution imaging. Over time, the tip can become contaminated or blunt, leading to inaccurate results. Maintaining a sharp, clean tip in operational conditions is challenging, and sometimes requires periodic tip renewal or treatment (e.g., scratching/heating to restore sharpness).
- **Environmental Sensitivity (Vibration Sensitivity) :** STM requires an extremely stable environment, as even minor vibrations can interfere with measurements. Laboratories typically require vibration isolation systems to minimize this issue.

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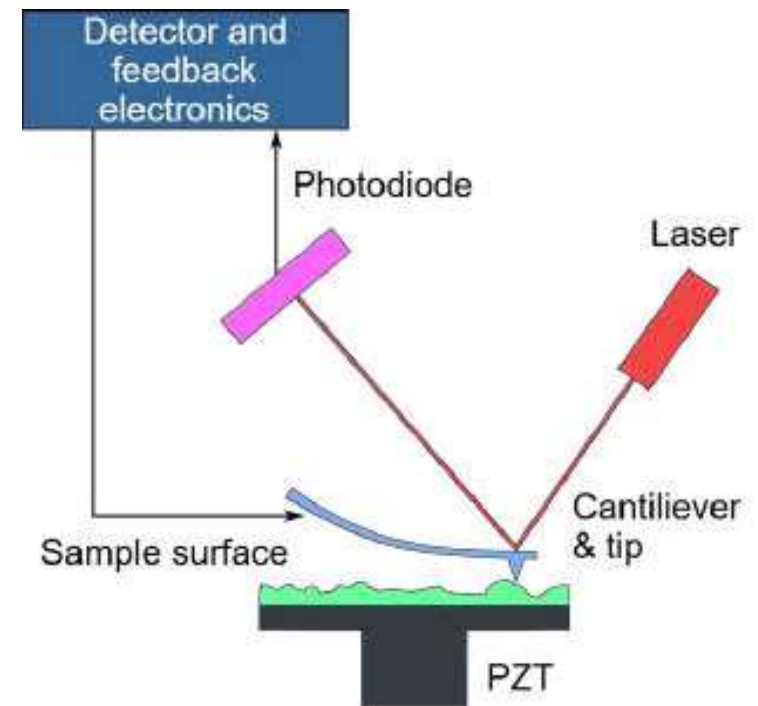
Atomic Force Microscopy

- Atomic Force Microscopy (AFM) is a **high-resolution scanning probe technique** capable of resolving surface features down to **sub-nanometer** scale. It works by **measuring the interaction forces** between a sharp probe (tip) and the sample surface.

Basic Components

AFM consists of:

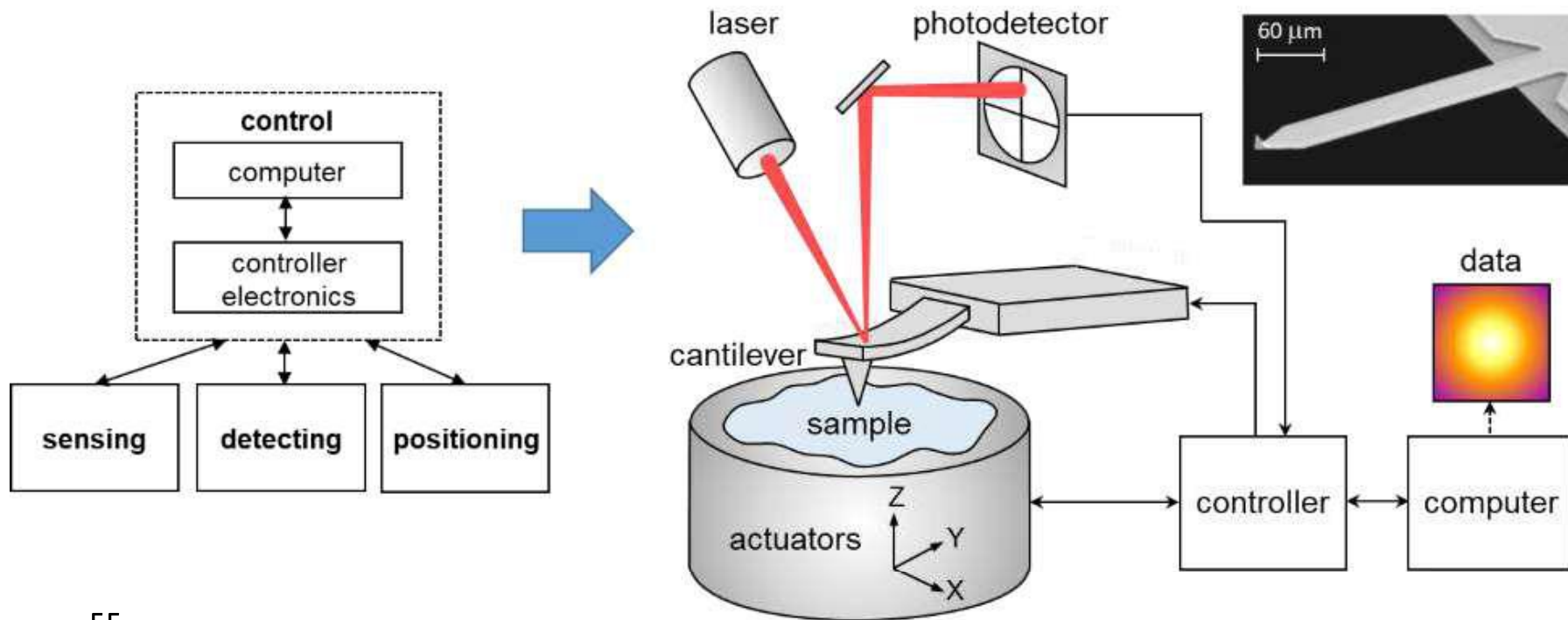
- Cantilever with a sharp tip**
 - Tip radius: 5–10 nm
 - Cantilever spring constant: 0.01–100 N/m
- Piezoelectric scanner**
 - Moves the sample (or tip) in x–y–z directions with Å-level precision.
- Laser and Position-Sensitive Photodiode (PSPD)**
 - To detect cantilever deflection.
- Feedback control electronics**
 - Keeps the interaction force constant.



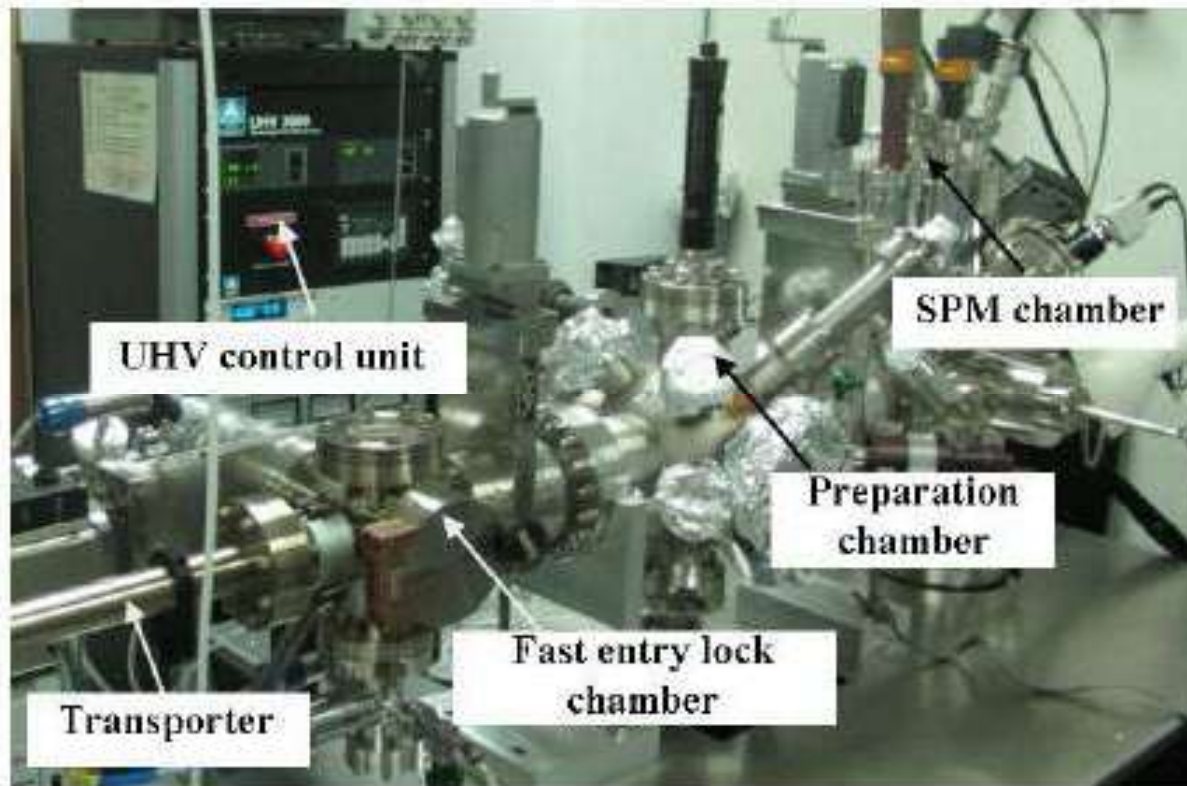
https://en.wikipedia.org/wiki/Atomic_force_microscopy

Atomic Force Microscopy

- Contact AFM: To acquire both topography and current maps
- Non-contact mode AFM: Topography
- Tapping mode-AFM: Topography

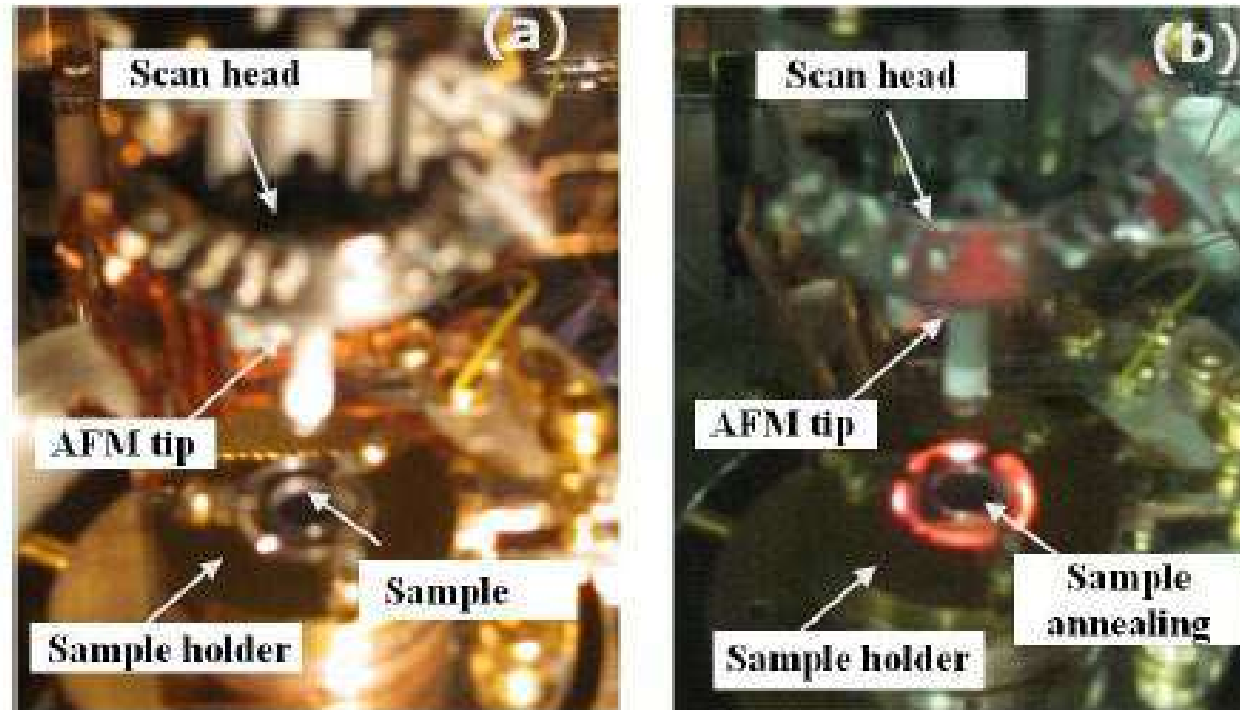


Atomic Force Microscopy



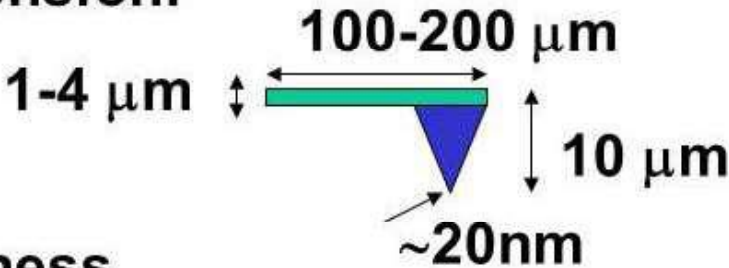
The UHV-SPM (RHK) system consisting of a fast entry lock chamber, a preparation chamber and SPM chamber.

Atomic Force Microscopy



(a) Image of the scan head assembly and sample holder, and (b) during sample annealing.

Atomic Force Microscopy

- **Material:** Silicon, Pt-Ir, Diamond, etc
- **Dimension:**

- **Stiffness**
 - soft: contact mode
 - stiff: vibrating (dynamic force) mode
- **Spring constant (k):**
0.1-10N/m
- **Resonance frequency:** 10~100kHz

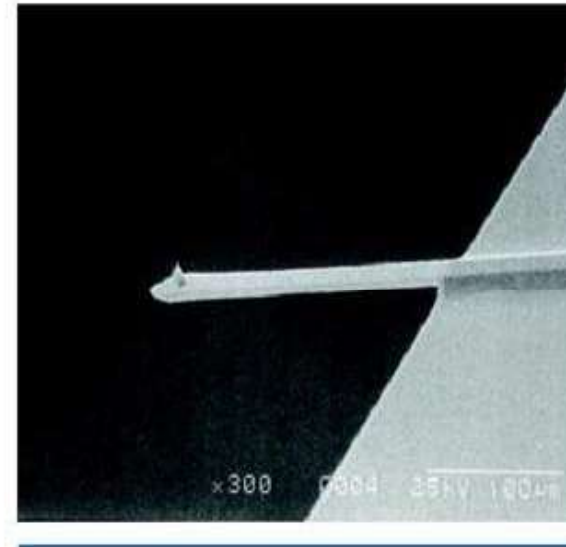


Figure 3. SEM image of an integrated single crystal silicon cantilever and tip which has an end radius of 2 to 10nm. Tips for AFM are typically made of silicon or silicon nitride. Bar=100μm.

Atomic Force Microscopy

Force

How are Forces Measured?

The probe is placed on the end of a cantilever (which one can think of as a spring). The amount of force between the probe and sample is dependant on the *spring constant* (stiffness) of the cantilever and the distance between the probe and the sample surface. This force can be described using Hooke's Law:

$$F = -k \cdot x$$

F = Force

k = spring constant

x = cantilever deflection

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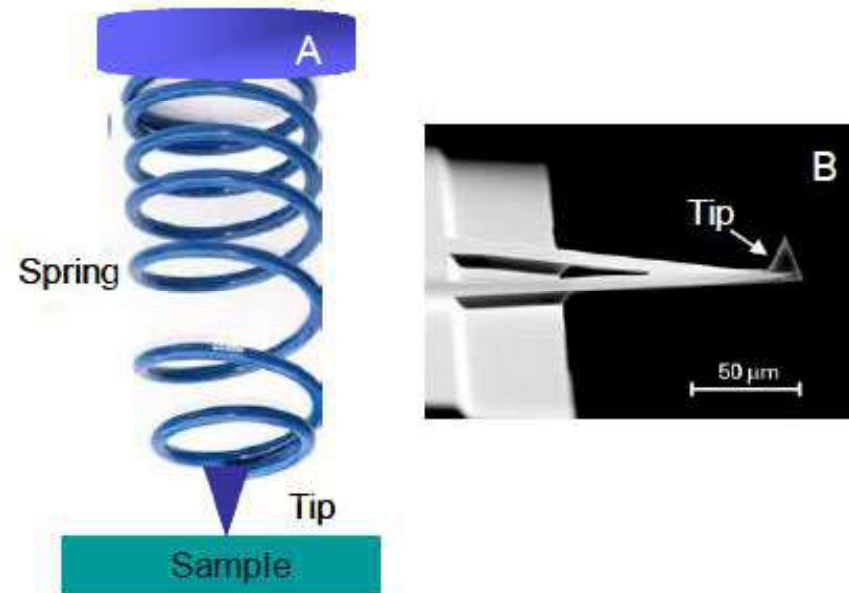


Figure 1. a) Spring depiction of cantilever b) SEM image of triangular SPM cantilever with probe (tip). (Image from [MikroMasch](#))

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Atomic Force Microscopy-Tip properties

The **spring constant (k)** of the **AFM tip's cantilever** plays a crucial role in both **contact** and **non-contact** modes, affecting sensitivity, resolution, and sample interaction.

1. Contact Mode and Spring Constant

- In **contact mode**, the tip remains in direct contact with the sample surface.
- The force applied by the tip follows **Hooke's Law**:
- $F = k \cdot d$ where **F** is the force, **k** is the spring constant of the cantilever, and **d** is the deflection.

Atomic Force Microscopy-Tip properties

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- $F = k \cdot d$ where **F** is the force, **k** is the spring constant of the cantilever, and **d** is the deflection.
- A **higher spring constant (stiffer cantilever)** is needed to avoid unwanted bending when scanning **hard materials** like metals and semiconductors.

Atomic Force Microscopy-Tip properties

2. Non-Contact Mode and Spring Constant

- In **non-contact mode**, the tip hovers just above the surface, detecting long-range forces (van der Waals, electrostatic, etc.) without touching.
- The cantilever oscillates at its **resonance frequency**, and **small force variations** shift this frequency.
 - A **low spring constant** is preferred, as a more flexible cantilever is more sensitive to small force changes.
 - Useful for **soft biological samples** to minimize sample damage.

Atomic Force Microscopy-Tip properties

Key Considerations:

Typical spring constant ranges:

Contact mode: $k \approx 0.1 - 50 \text{ N/m}$ (depends on material hardness).

Non-contact mode: $k \approx 1 - 10 \text{ N/m}$ (enough stiffness to maintain oscillation).

- In **contact mode**, the spring constant determines the applied force and potential sample damage.
- In **non-contact mode**, it affects oscillation stability and sensitivity to long-range forces.

Choosing the right spring constant is critical for **minimizing damage, maximizing resolution, and obtaining accurate measurements.**

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Atomic Force Microscopy

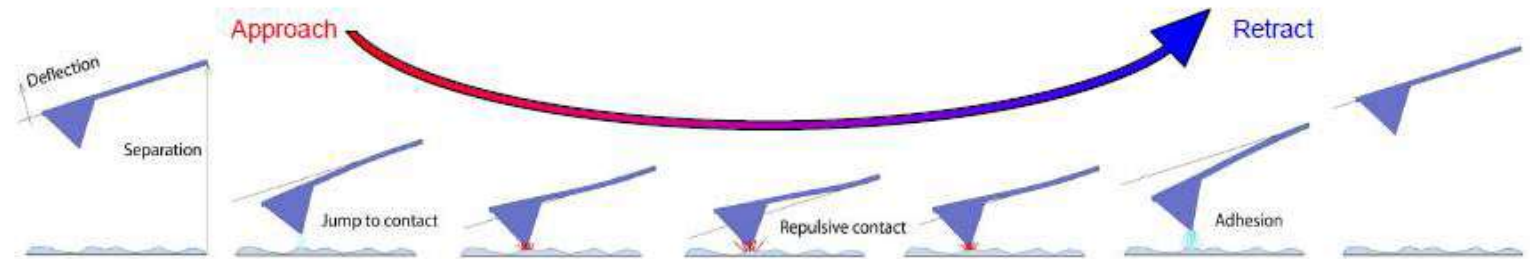
Force –distance curve



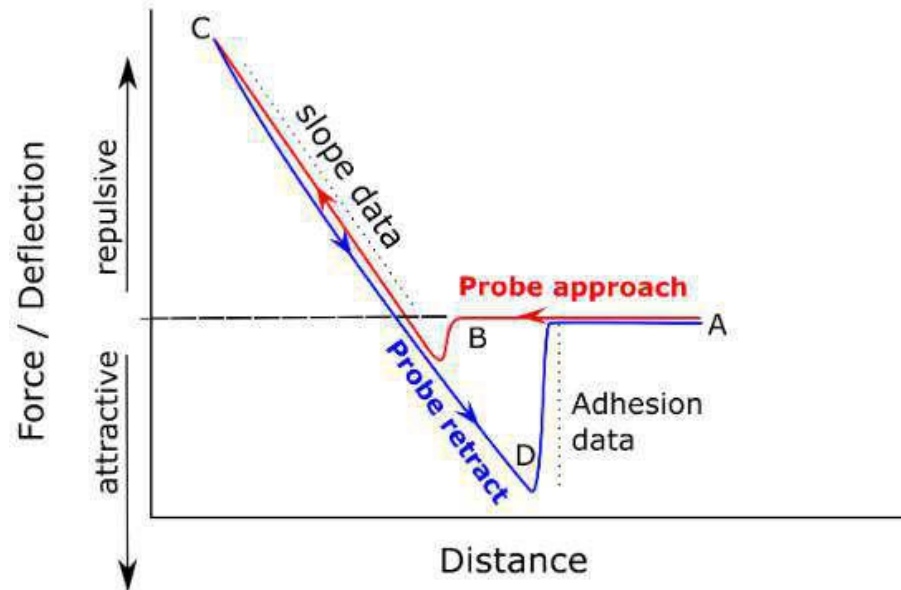
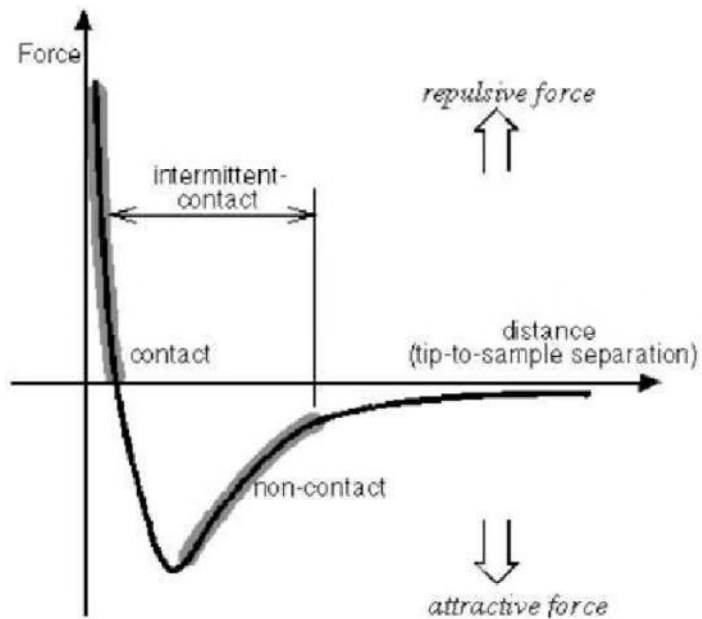
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Atomic Force Microscopy

Force –distance curve



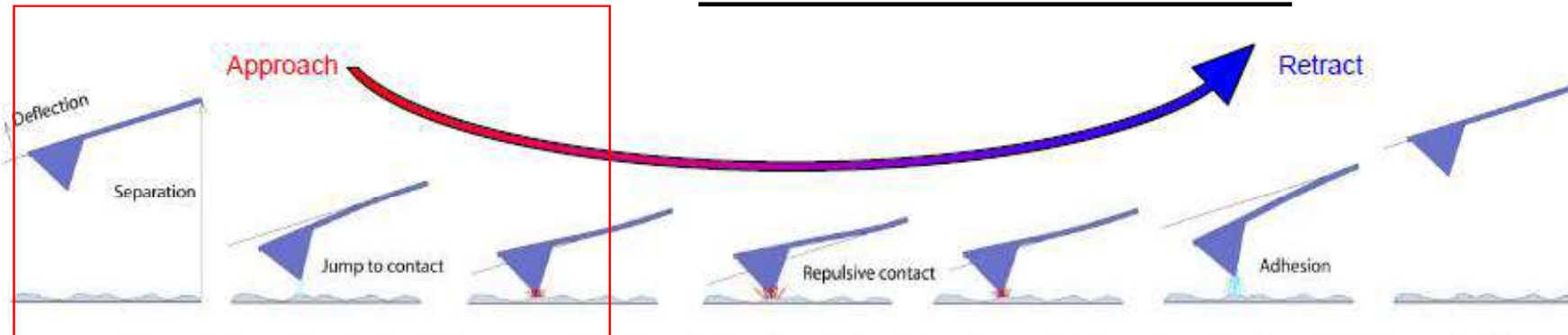
Schematic diagram of the vertical tip movement during the approach and retract parts of a force spectroscopy experiment.



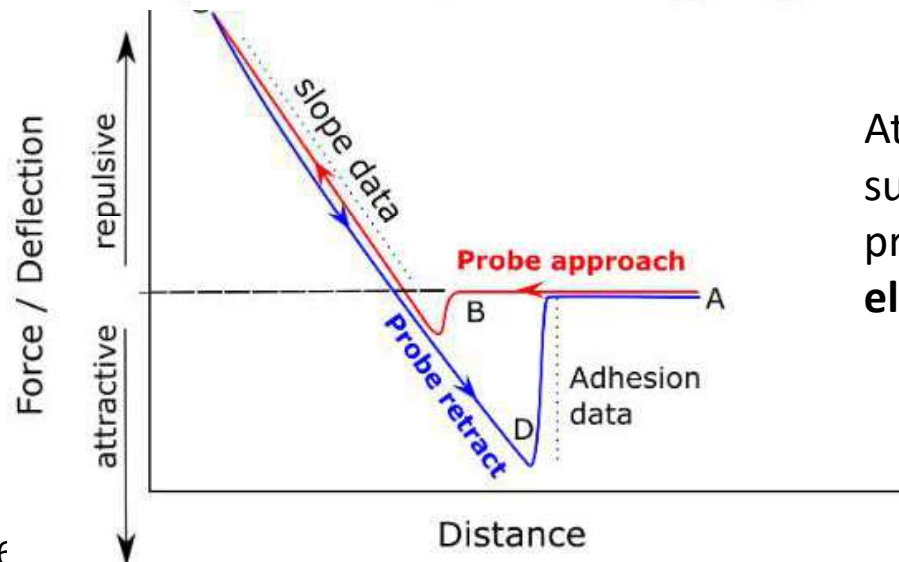
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Atomic Force Microscopy

Force –distance curve



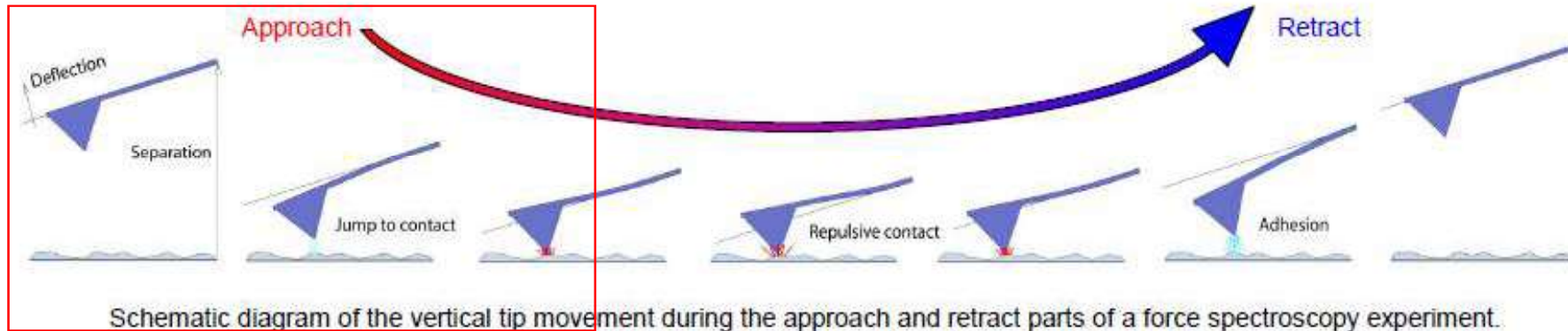
Schematic diagram of the vertical tip movement during the approach and retract parts of a force spectroscopy experiment.



Attractive forces dominate when the tip is far from the surface but still within interaction range. These are primarily **van der Waals forces** and occasionally **electrostatic forces** if the surface or tip is charged.

Atomic Force Microscopy

Force –distance curve

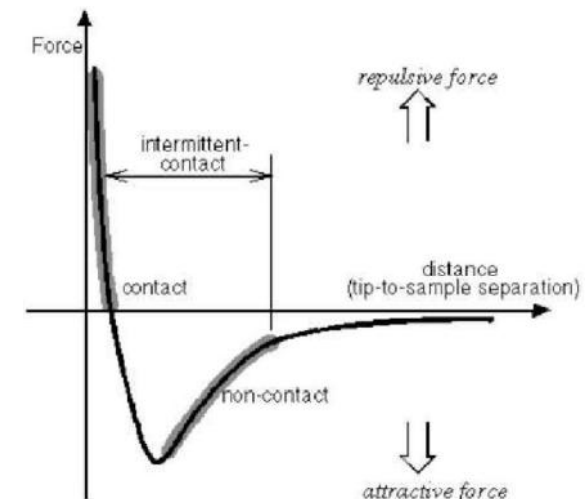


What Are Van der Waals Forces?

Van der Waals (vdW) forces are **weak, attractive intermolecular forces** arising from **electric dipole interactions** between atoms or molecules. Even when a material is electrically neutral, random electron fluctuations create **temporary dipoles**.

These temporary dipoles induce dipoles in nearby atoms → resulting in **attractive force**.

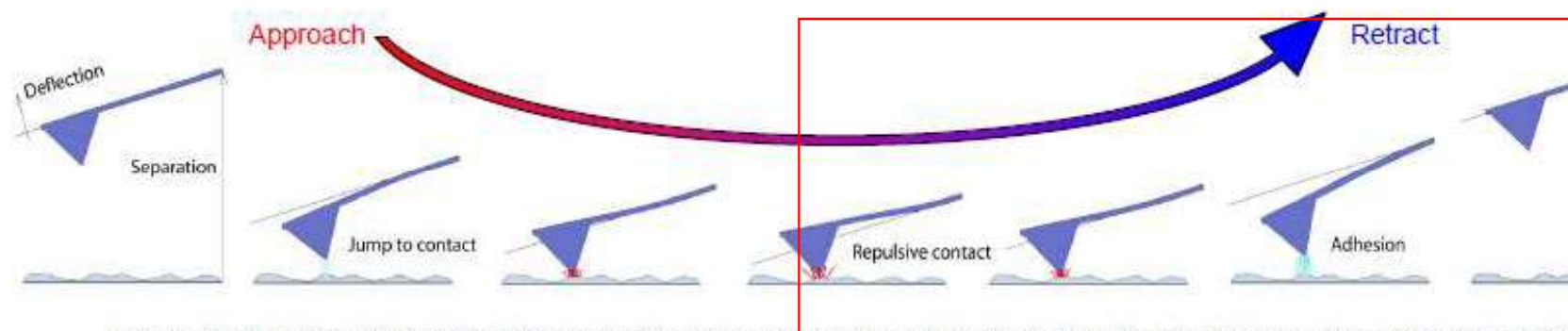
AFM exploits this attraction between the **tip atoms** and **sample atoms**.



Atomic Force Microscopy

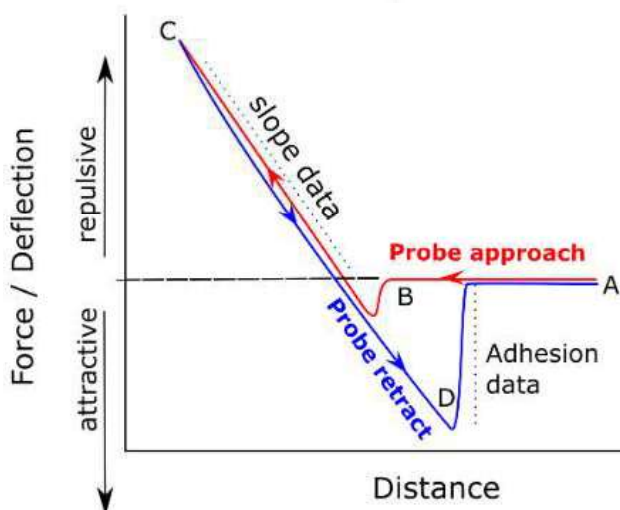
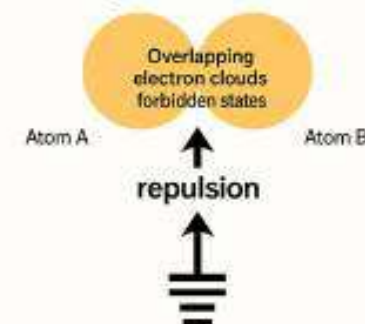
Force –distance curve

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Schematic diagram of the vertical tip movement during the approach and retract parts of a force spectroscopy experiment

Short-Range Pauli Repulsive Force



- Repulsive forces dominate when the tip is extremely close to or in contact with the surface. These arise from **Pauli exclusion principle** and **electron cloud overlap**.
- At very short distances, the electron clouds of atoms on the tip and surface start to overlap, creating a strong repulsive interaction.
- The repulsion causes the cantilever to bend away from the surface, resulting in an upward deflection in the graph.

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Atomic Force Microscopy

Applications

Adv:

- Easy sample preparation
- Non-destructive imaging
- Accurate height information
- Works in vacuum, air, and liquids
- Sample not required to be conductive

Polymers, ceramics, glass

Metals

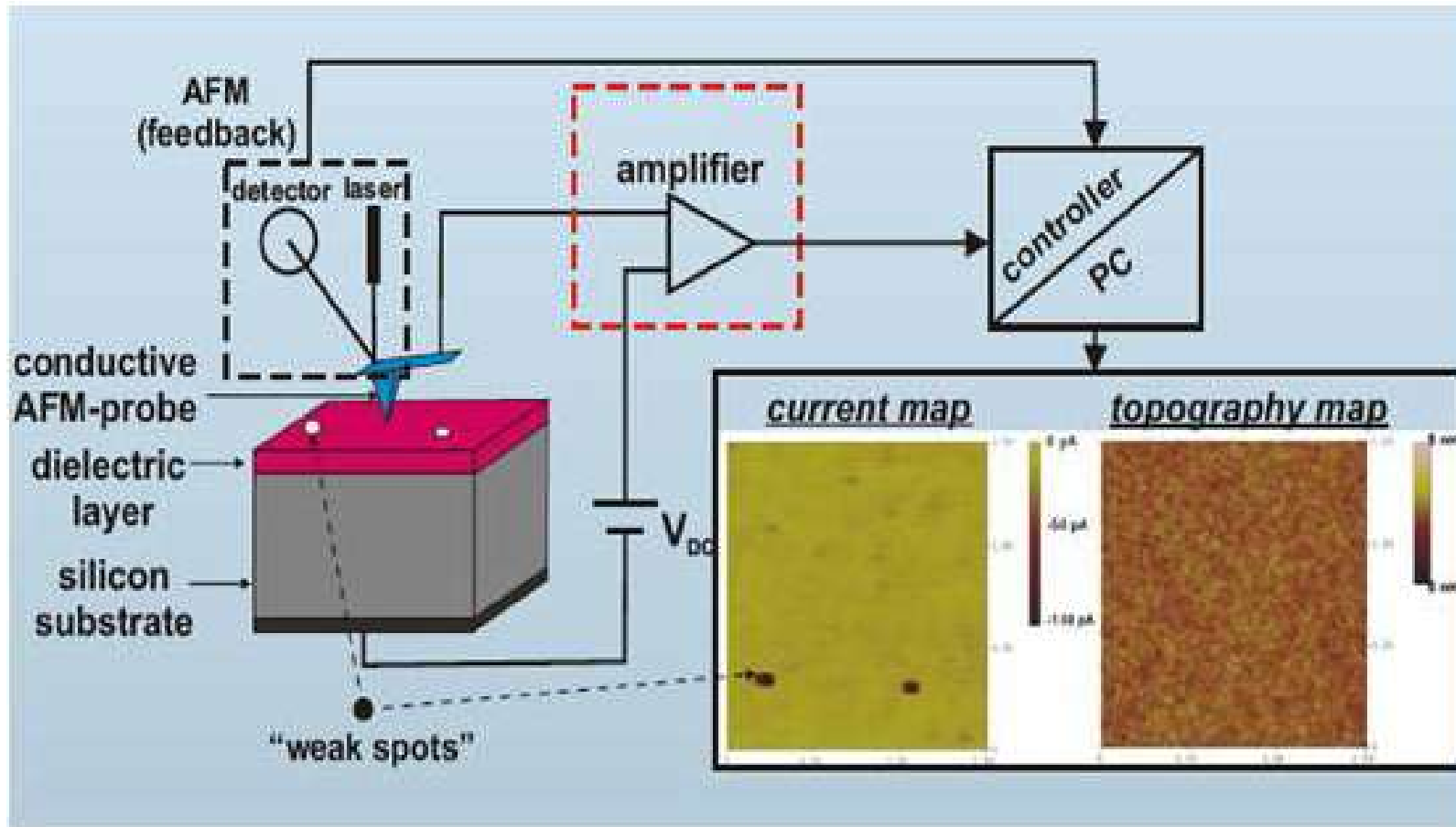
Biological samples

Dis-adv:

- Limited vertical range
- Limited magnification range
- Highly Dependent on AFM probes.
- Tip or sample can be damaged

Atomic Force Microscopy

- **Contact AFM**: Topography and current map of thin Dielectric materials (Conductive AFM)



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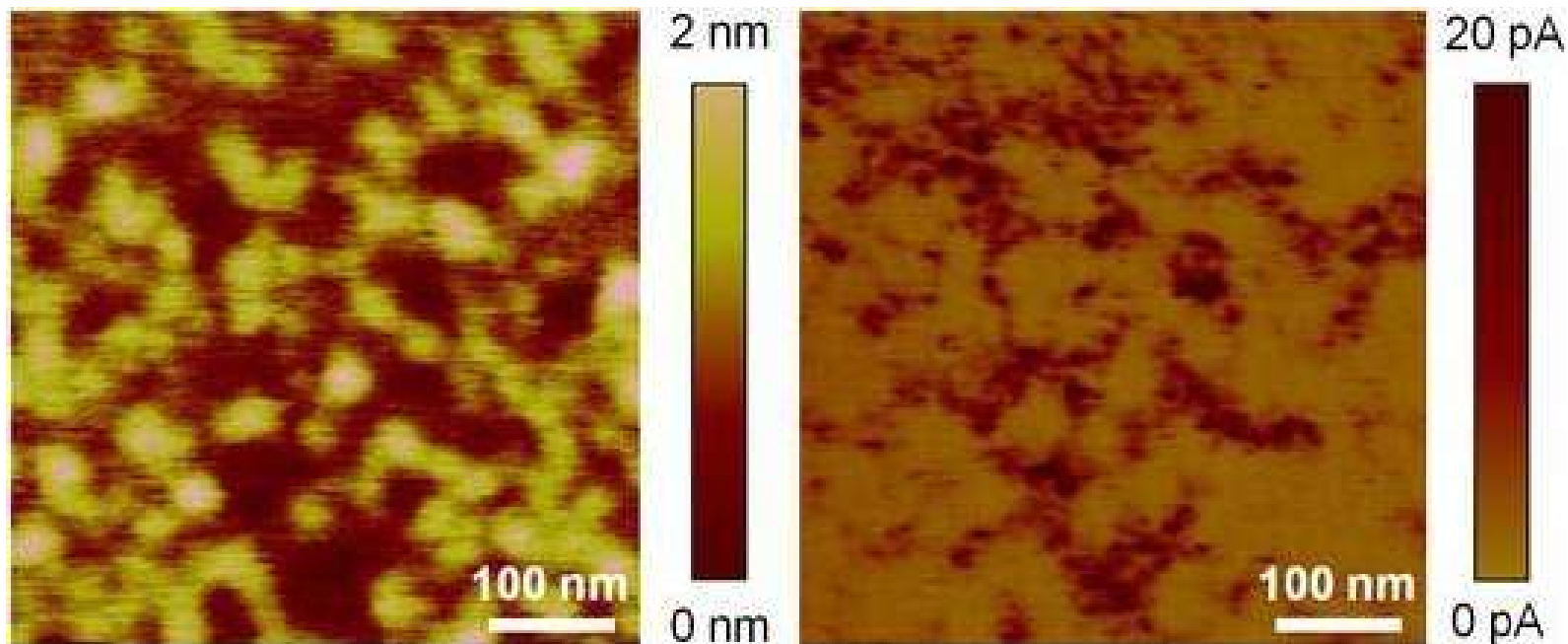
Atomic Force Microscopy



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Atomic Force Microscopy

- Contact AFM: Topography and current map of thin Dielectric materials (Conductive AFM)

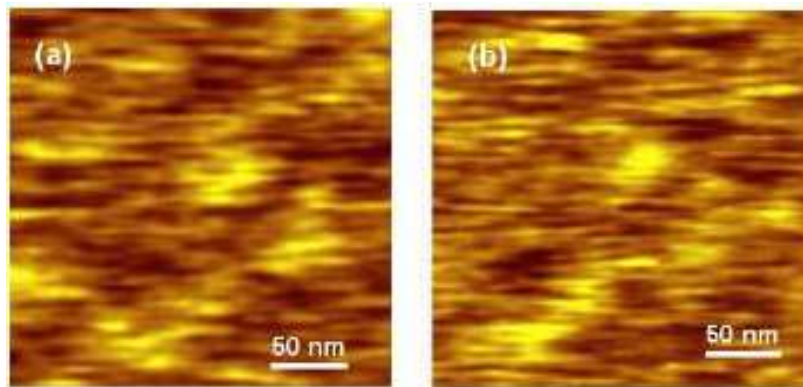


Topographic (left) and current (right) maps collected with CAFM on a polycrystalline HfO_2 stack. The images show very good spatial correlation.

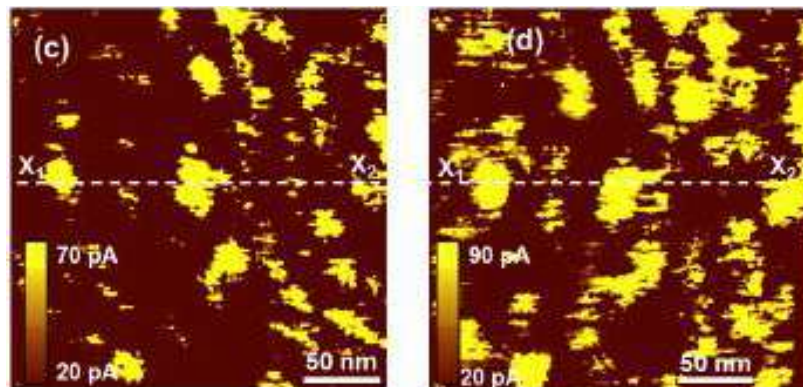
Dilation of leakage spot upon stressing -CAFM

- Larger area scanning: Dilation of leakage spot with Uniform stressing (+6V)

Topography



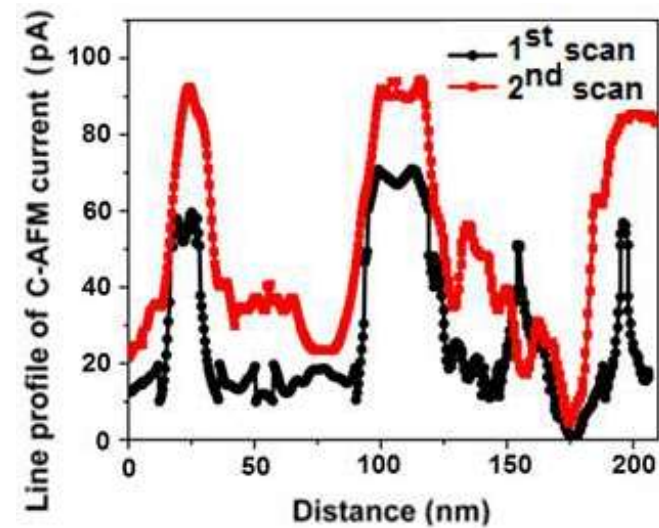
Current map



First Scan

Second Scan

Along X_1 - X_2



Atomic Force Microscopy

- **Non-contact AFM**: Topography at high resolution for sensitive samples

Contact mode

Contact mode



Noncontact mode

Noncontact mode



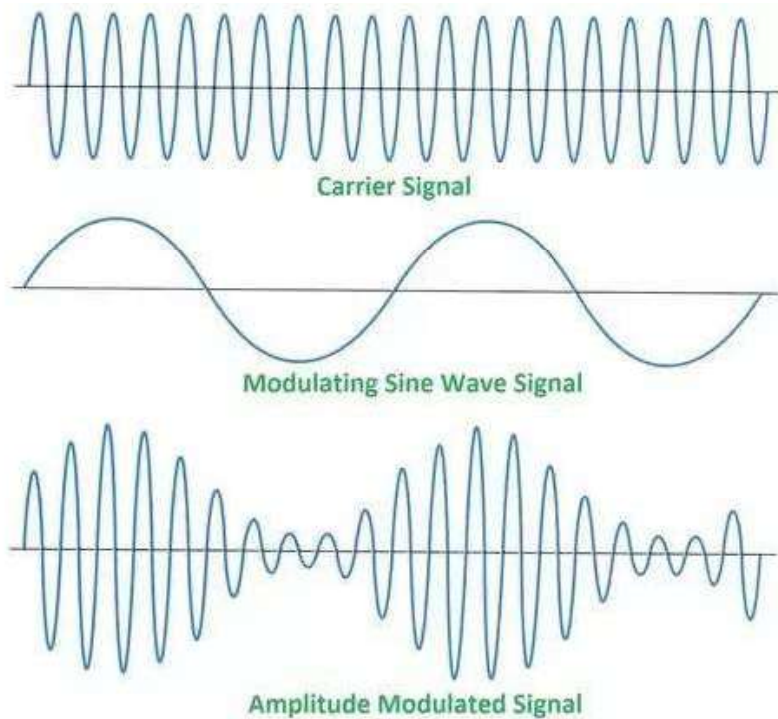
The two most common modes of nc-AFM operation,

- Amplitude modulation (AM),
- Frequency modulation (FM)

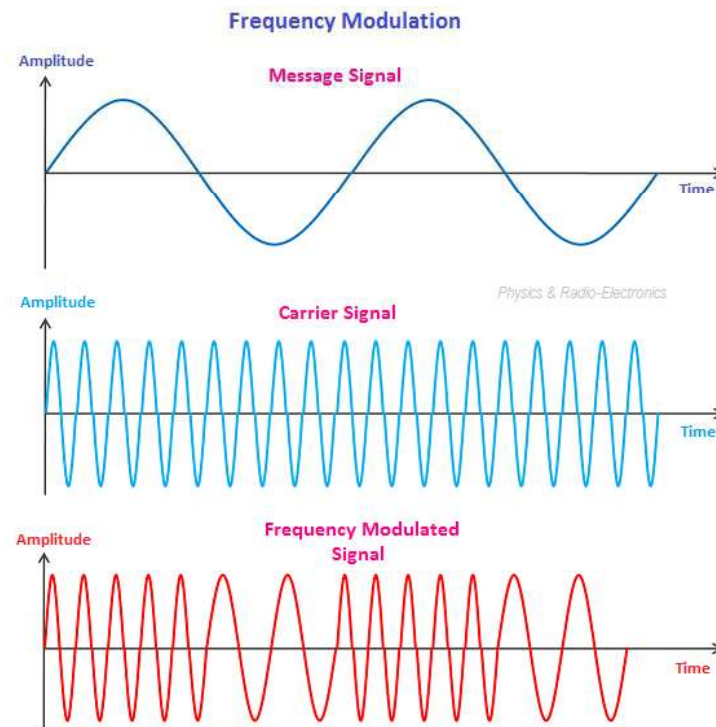
Atomic Force Microscopy

- Non-contact AFM:

Amplitude modulation (AM)



Frequency modulation (FM)



Atomic Force Microscopy

- Non-contact AFM:

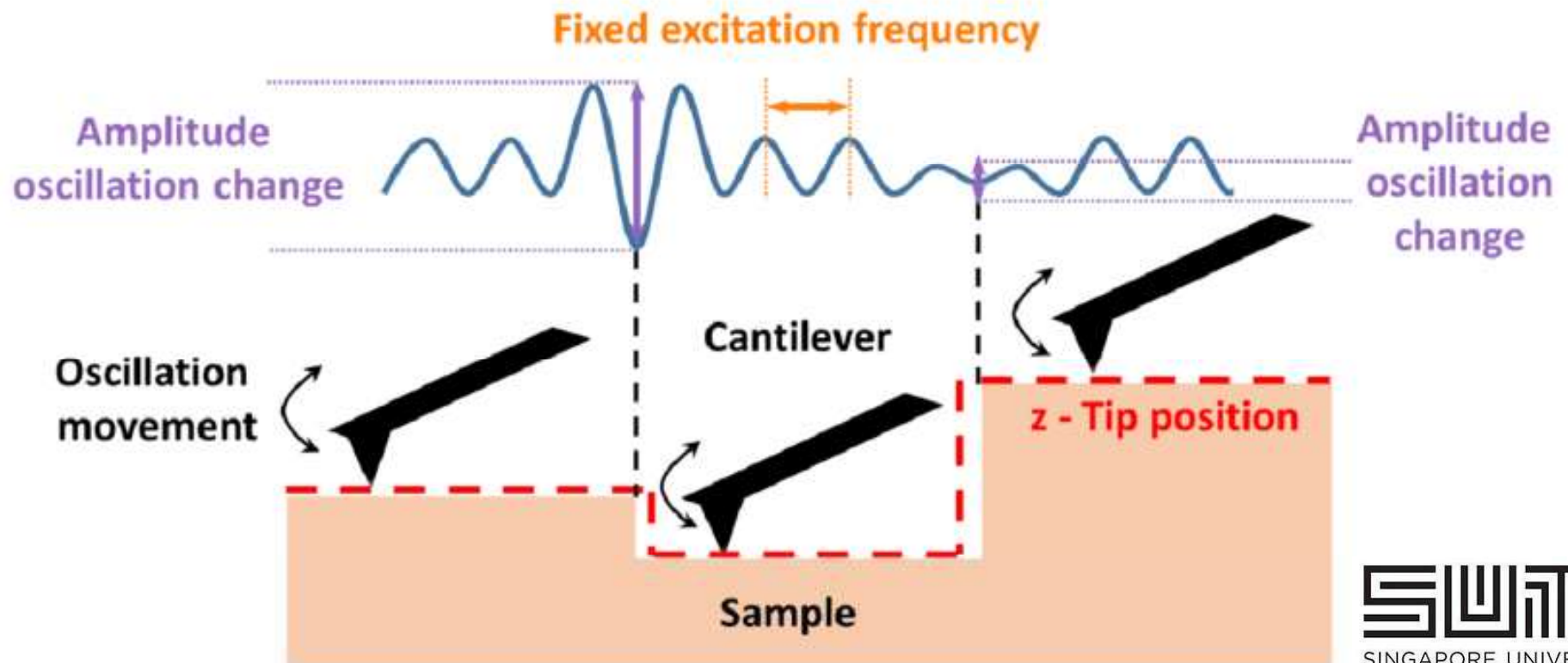
- oscillating cantilever, tip not in contact with sample
- used for soft samples
- imaging in vacuum

Non-contact imaging (center) generally provides low resolution and can also be hampered by the contaminant (e.g., water) layer which can interfere with oscillation.

Atomic Force Microscopy

- Non-contact AFM:

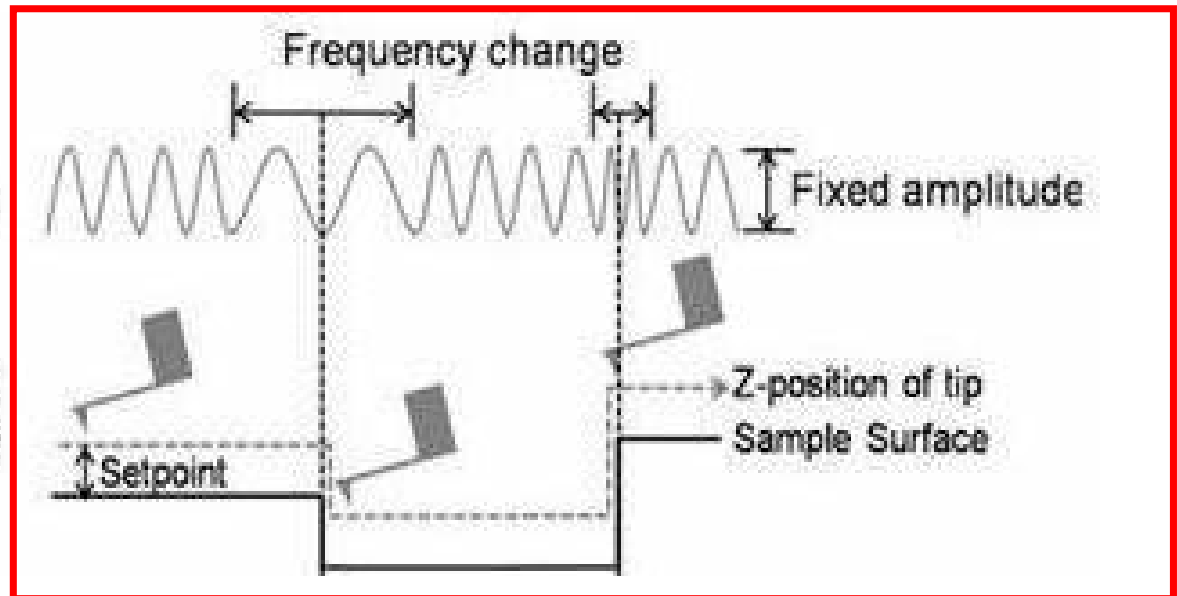
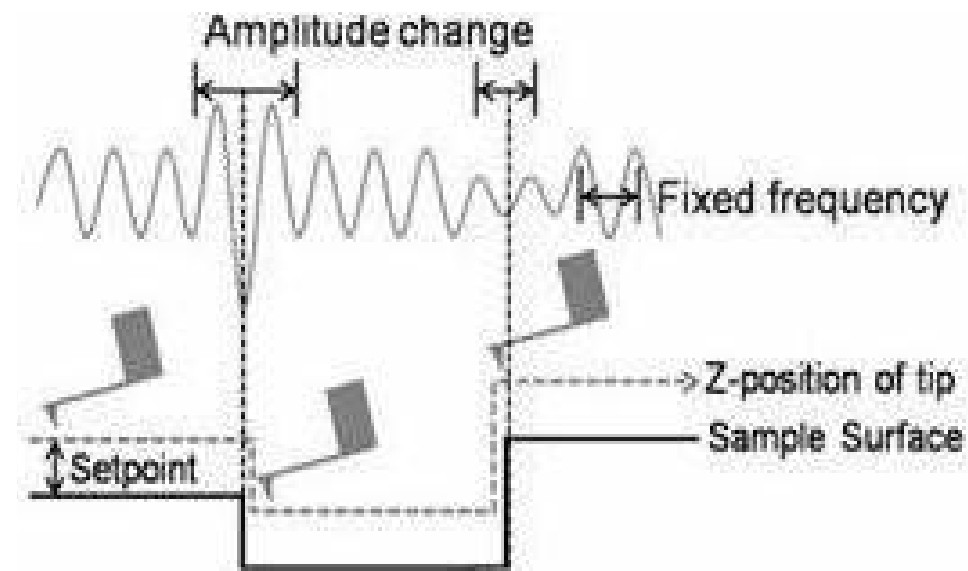
Amplitude modulation (AM)



Atomic Force Microscopy

- Non-contact AFM:

Frequency modulation (FM)



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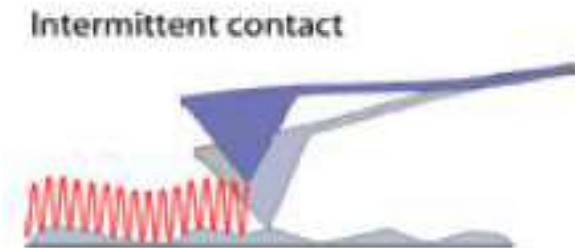
Atomic Force Microscopy



Atomic Force Microscopy

• Tapping Mode AFM:

- oscillating cantilever, tip touching surface gently and frequently
- often used for biological samples
- imaging in air and liquid
- good resolution



Atomic Force Microscopy

AFM Modes of Operation	Working Principle	Advantage	Disadvantage
Contact Mode	Physical contact between the tip and the surface	High scan speeds	- Damage to soft sample - Later forces may produce image artefacts
Non-contact Mode	No contact between the tip and the sample	- High resolution - No damage to sample	Slower scan speed if compared with both contact and tapping mode
Tapping Mode	Intermittent and short contact between the sample and the tip	Minimal damage to sample	Slower scan speed if compared with contact mode